KRISTIAN HAJDARI - UX DESIGN STUDENT - 2026

Tactile Interaction and Visitor Engagement within the being human exhibition at the Welcome Collection

Exploring how button based artefacts in particular influence participation, behaviour, and engagement within exhibition spaces.

Initial Curiosity and Topic Development

After reading the project brief during weeks 1 and 2, I visited the Being Human exhibition at the Wellcome Collection to begin understanding the space through direct observation and personal experience.

At first, I was broadly interested in trust within exhibitions and how interactive environments shape visitor trust and engagement.
My initial idea was to focus specifically on students as a target audience. I wanted to investigate how younger visitors engaged with interactive exhibitions and whether identity based content influenced their participation differently.

During my first observations, I started noticing strong behavioural differences between interactive and non interactive artefacts. Certain installations consistently encouraged longer engagement, experimentation and social interaction.

These early observations became the starting point for the project, although my initial research direction later changed significantly through feedback and continued observation.

Early Research Direction and Initial Problems

At the beginning of the project, my research question was still very broad and focused on trust, interaction and visitor engagement in general, i also attempted to narrow the audience specifically towards students.

However, after tutor feedback, I realised there were major problems with this approach and i realized that i could not accurately confirm whether the visitors I observed were actually students without directly asking them.
This created a methodological issue within the research process.

At the same time, my research question was still covering too many themes at once:

- identity
- participation
- engagement
- exhibition interaction
- visitor behaviour

Because of this, the investigation lacked a clear and focused direction.

Returning Back and Refining the Focus

After reflecting on the feedback, I decided to step back and reconsider the focus of the project.

Instead of concentrating on a specific demographic group, I shifted the research towards observable behaviours that could be consistently analysed through field observation, and this changed the direction of my project significantly.

Rather than asking “How do students engage with identity based exhibitions?”

I began focusing more on:

- tactile interaction
- participation
- experimentation
- repeated behaviour
- visitor engagement within the exhibition space

This moment became an important turning point within the process because it helped narrow the project into something more researchable, observable and evidence based.

First Observations Inside the Exhibition

During repeated visits to the exhibition, two artefacts immediately stood out to me: the Invisible Woman and the Jukebox.

The Invisible Woman attracted attention almost instantly because of its location at the entrance and its tactile interaction system. Visitors repeatedly pressed buttons, tested different combinations and stayed engaged for longer periods of time.

The Jukebox created a more social form of participation. Visitors gathered around the artefact, selected songs together, laughed, danced and reacted collectively to the experience.

Compared to other “passive” artefacts at the exhibition, these artefacts consistently generated:

- longer dwell times
- curiosity
- experimentation
- social interaction
- repeated engagement

At this point, tactile participation became the strongest recurring behaviour across my observations.

From Identity to Tactile Engagement

At the beginning of the project, trust was the central focus of the investigation.

However, through repeated observations, I realised that the most noticeable behavioural patterns were not only connected to exhibition content, but to the way visitors physically interacted with artefacts and with each other.

Interactive installations repeatedly encouraged:

- active participation
- “replay” behaviour
- exploration
- collaborative interaction
- longer engagement

Because of this, the project gradually shifted away from identity as a broad theme and moved towards tactile engagement and visitor participation within exhibition environments.

Developing the Research Question

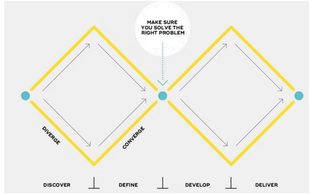
After refining the project direction, I began focusing more specifically on how tactile interaction influenced visitor engagement inside the exhibition.

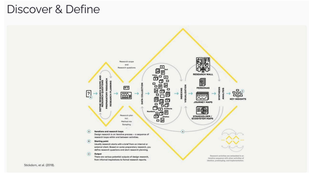
The repeated behaviours I observed around interactive artefacts led me towards my final research direction and this eventually developed into my main research question:

How do tactile interactions with button based artefacts affect visitor engagement within the exhibition space?

I wanted to better understand how physical interaction, especially through button based artefacts, shapes behaviour, curiosity and participation and engagement within museum spaces.

Double Diamond Framework





This process strongly connected with the first stages of the Double Diamond framework, which are discover and define.

An important part of the project involved going backwards, refining the focus and reconsidering earlier assumptions before moving forwards again.

Instead of immediately developing solutions, I focused on:

- observation
- behavioural analysis
- identifying recurring patterns
- refining the research direction
- synthesising findings

Why Secondary Research Became Important

After refining my research question, I started doing secondary research to better understand how interaction influences engagement within exhibition environments.

At the beginning of the project, many of my ideas were still broad and unfocused. I was interested in identity, trust, interaction and visitor experience all at the same time, but I needed stronger contextual understanding before continuing with primary research.

O'Leary (2021) explains that secondary research helps researchers understand what is already known, identify gaps in knowledge and position their own investigation within wider discussions. This became important while narrowing my topic and refining the direction of the project.

Rather than immediately jumping into conclusions, I used secondary research to better understand:

- button based interaction
- visitor controlled interaction
- observational research methods

Photo took from The New York Times website, not taken by me

Photo took from The New York Times website, not taken by me

Beginning the Research Process

During the early stages of the project, I started collecting information from:

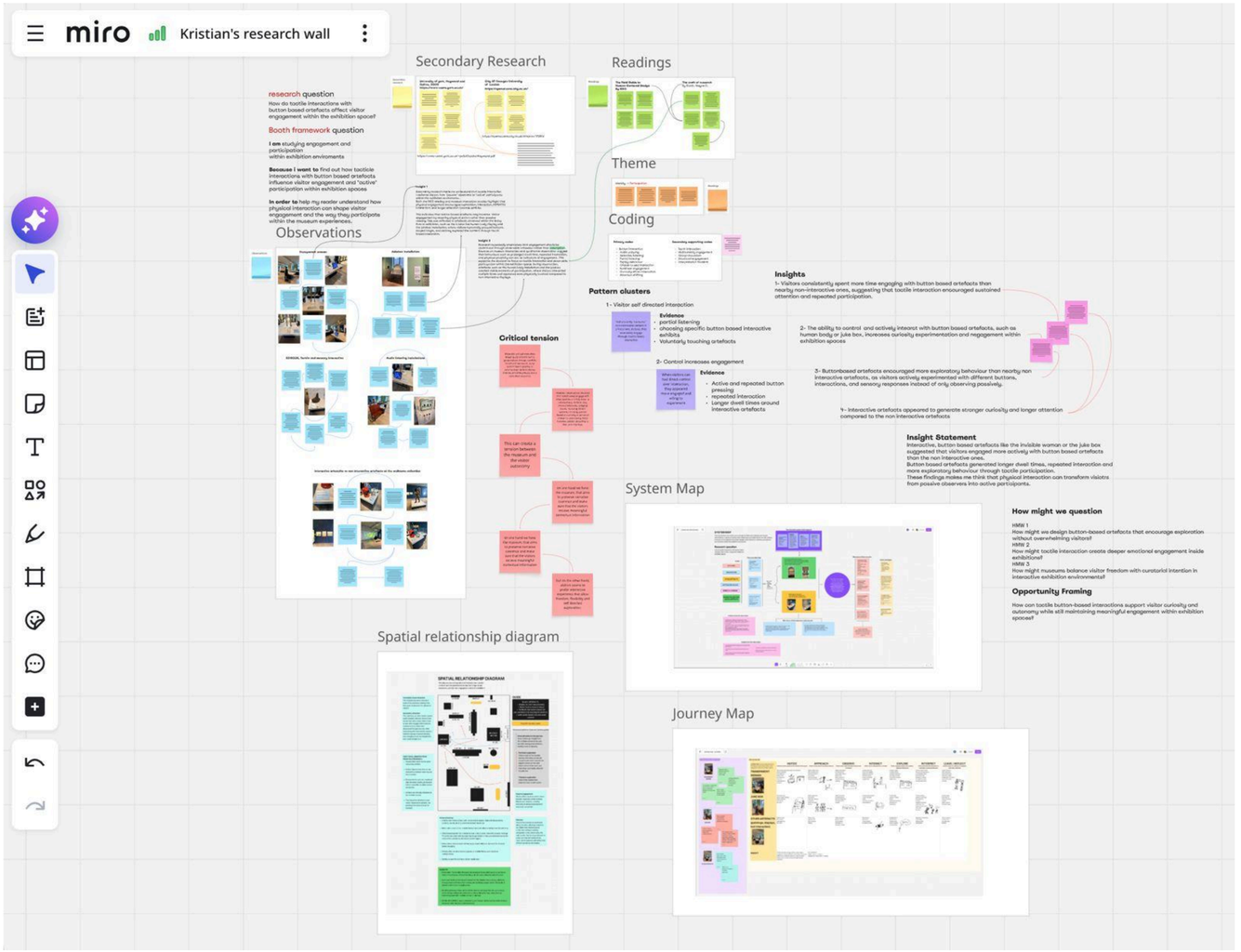
- academic readings
- museum interaction studies
- design research books
- online articles
- observational notes
- exhibition visits

At first, my research wall contained broad themes connected to trust, participation and visitor engagement. However, after repeated observations inside the exhibition space, tactile button based interaction started appearing more consistently across both my primary and secondary research.

As I continued researching interactive museum environments, I repeatedly found connections between:

- physical button interaction
- visitor control
- replay behaviour
- experimentation
- curiosity
- active engagement

Large screenshot of my research wall



Trust → Identity → Participation

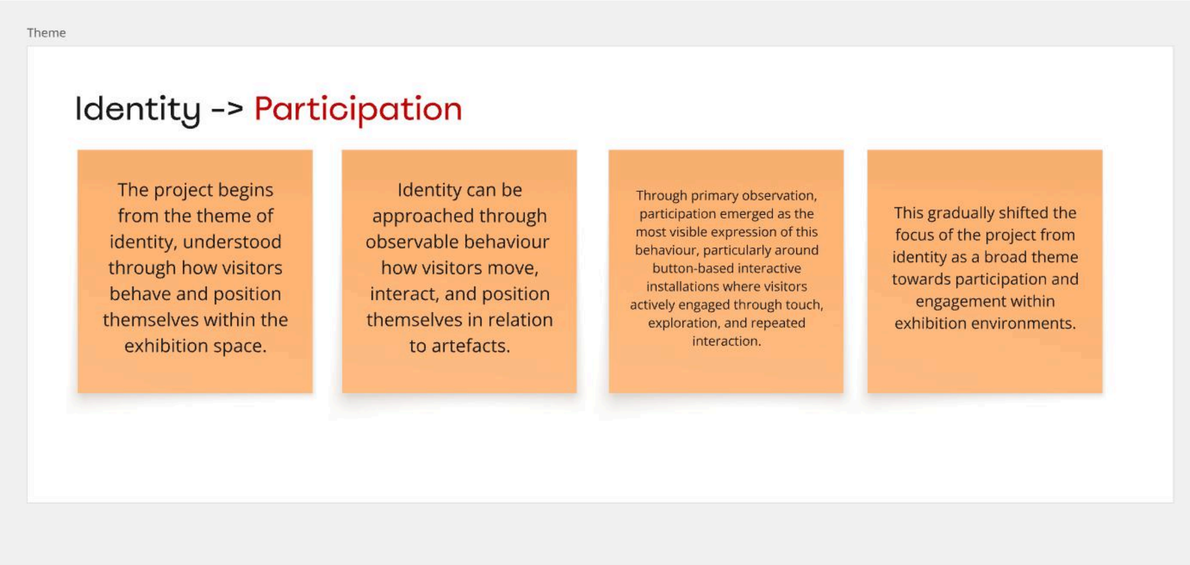
At the beginning of the project, my investigation originally started around the theme of trust within exhibition environments. I was interested in how visitors emotionally respond to spaces, artefacts and institutional environments, particularly within museums where information, authority and interpretation influence visitor perception.

However, after early observations and research, I realised that trust itself was difficult to observe directly through behaviour alone. Instead, I started noticing how visitors physically positioned themselves, interacted with exhibits and behaved within the exhibition space.

This gradually shifted the project towards identity, where I explored how identity could be understood through observable behaviour, movement and interaction inside the exhibition environment.

As my observations developed further, participation emerged as the strongest visible expression of this behaviour, particularly around button based interactive artefacts where visitors actively engaged through touch, exploration, replay interaction and physical control.

The Invisible Woman and the Jukebox became especially important because visitors were not simply observing them passively. Instead, they repeatedly pressed buttons, experimented with controls and interacted socially around the artefacts, making participation far more visible and measurable than broader themes such as trust or identity alone.



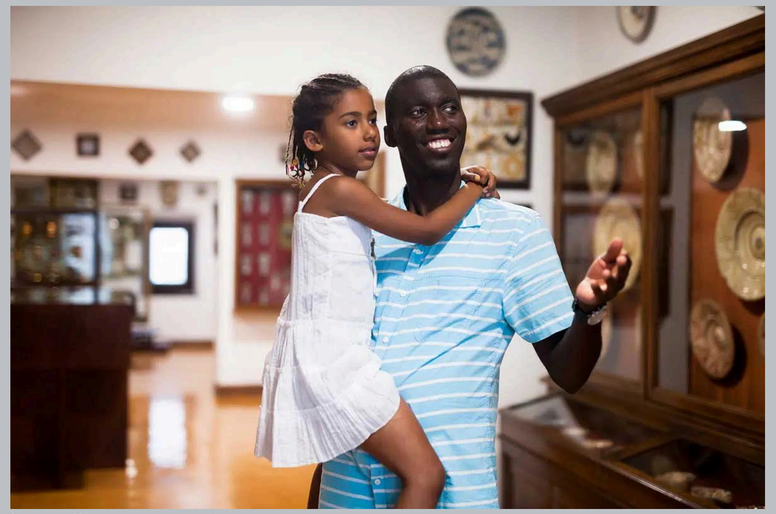
Screenshot taken from my research wall

Curiosity Through Observable Behaviour

While researching how engagement and curiosity can be identified within museum environments, I found Stephen Bitgood's Attention and Value: Keys to Understanding Museum Visitors (2013).

The text explains that visitor engagement can often be observed through visible behaviours such as:

- prolonged attention
- stopping behaviour
- repeated engagement
- physical orientation towards exhibits



Images ffrom museumnext

This became highly relevant to my own observations because many visitors interacting with button based artefacts displayed these exact behaviours repeatedly throughout the exhibition.

For example, visitors frequently remained longer around the Invisible Woman installation, replayed button interactions multiple times, and physically repositioned themselves around the artefact while exploring different outcomes.

Images ffrom museumnext

Participation Through Button Based Interaction

As my research developed further, I became increasingly interested in how button based artefacts encourage participation within exhibition spaces.

Helen Armstrong explains that participatory design shifts audiences away from passive consumption and towards active contribution, where users become part of the experience itself rather than simply receiving information.

This strongly reflected what I was observing inside the Being Human exhibition. Artefacts such as the Invisible Woman and the Jukebox encouraged visitors to physically participate through pressing buttons, testing controls and replaying interactions multiple times.

Haywood and Cairns (2005) also describe engagement within interactive museum exhibits through participation and direct manipulation, explaining that visitors actively interact with systems in order to shape their own experience.

This became highly relevant to my investigation because button based artefacts allowed visitors to directly control the interaction through tactile input and immediate feedback.

image from researchgate

Tactile Feedback, Curiosity and Replay Behaviour

As I continued researching interactive exhibitions, I became more specifically interested in how button based interaction systems influence visitor curiosity and engagement.

Ntalla (2017) explains that responsive and intuitive interaction plays an important role within interactive museum experiences because visitors engage more comfortably when interactions feel predictable and responsive to their actions.

This strongly reflected what I observed around button based artefacts within the Being Human exhibition.

Visitors frequently:

- pressed buttons repeatedly
- experimented with different combinations
- replayed interactions multiple times
- tested different outcomes
- stayed longer within the space
- observed others before participating themselves

The immediate feedback generated through pressing buttons appeared to encourage experimentation, curiosity and prolonged engagement.

Similarly, research discussing interactive museum technology suggests that visitor controlled systems can positively influence participation and engagement by creating more immersive and active experiences.

Learning Through Observation

Alongside museum interaction research, I also explored readings connected to observational research and human centred design methods.

IDEO's Field Guide to Human-Centred Design explains that researchers cannot design for people without understanding their perspective first. This became important within my own process because it encouraged me to personally experience the exhibition before analysing visitor behaviour.

Instead of relying only on assumptions, I wanted to understand how engagement naturally emerged through thing like,button pressing interaction, movement, replay beahviour etc..

Stickdorn et al. (2018) describe preparatory research as a process of building contextual understanding before entering fieldwork. This strongly reflected the way I used secondary research to shape my observational approach before continuing with primary research inside the exhibition.

Building the Research Wall

As both primary and secondary research developed, I began organising observations, readings and behavioural patterns through a research wall.

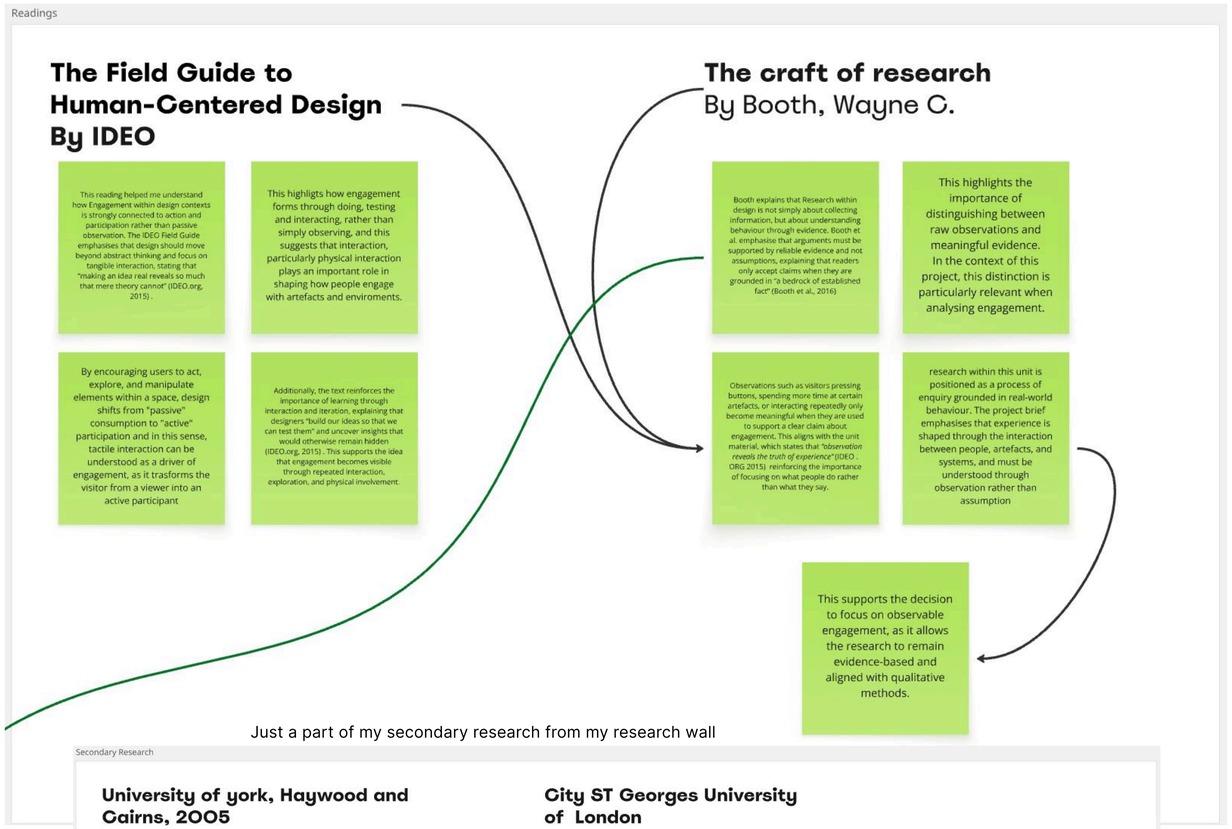
The wall allowed me to visually connect:

- button based interaction
- tactile participation
- replay behaviour
- visitor control
- curiosity
- social interaction
- engagement within the exhibition space

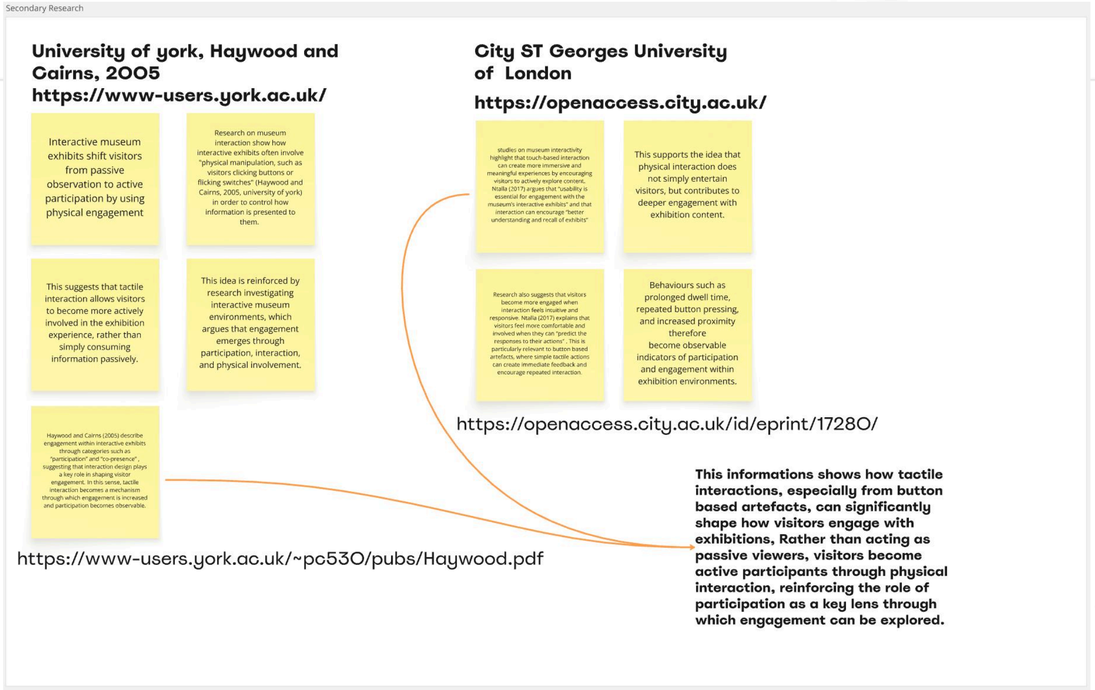
Instead of simply collecting information, the research wall helped reveal recurring relationships between physical interaction and visitor behaviour.

analysis identifies what happened, while synthesis helps explain what the findings actually mean, Youngblood et al. (2020) argue that synthesis is not about summarising data, but transforming it into insight.

This became an important part of my process because the research wall helped move the project beyond simple observation and towards a clearer understanding of how tactile button based interaction shapes visitor engagement.



Just a part of my secondary research from my research wall

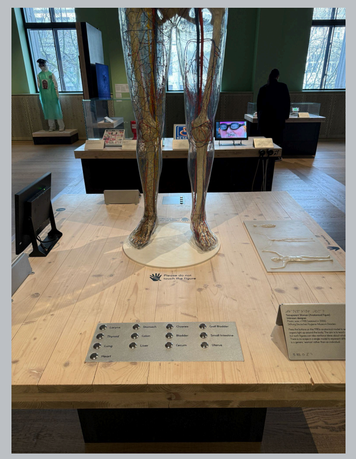


Connecting Secondary Research...

Across multiple readings and sources, button based tactile interaction repeatedly appeared as a key factor influencing visitor engagement within exhibition environments, themes such as tactile participation, visitor control, physical interaction and active engagement consistently reflected the same behaviours I had already been observing at the Wellcome Collection.

This became important because the secondary research did not simply provide theoretical information.

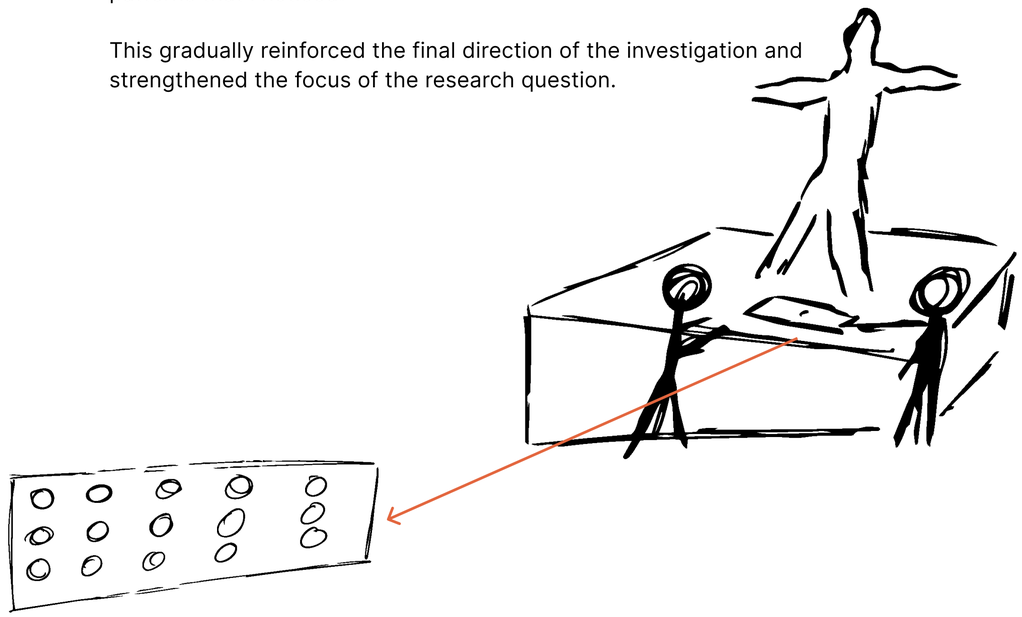
Instead, it helped validate and support the behavioural patterns emerging through my own field observations at the wellcome collection, in particular around the Invisible Woman and the Jukebox artefacts.



...to My Investigation

As the project developed further, tactile interaction with button based artefacts became the strongest recurring connection across my observations, my secondary research and the visitor behaviour patterns that i noticed.

This gradually reinforced the final direction of the investigation and strengthened the focus of the research question.



Understanding interactive exhibition behaviour

At first, my project explored the theme of trust within exhibition environments. While researching this area, I realised that trust was often connected to how visitors emotionally and physically engaged with spaces, artefacts, and systems around them.

This gradually shifted my focus towards identity, particularly how visitors express themselves through movement, behaviour, and interaction within exhibitions.

However, after several visits to the Being Human exhibition, another pattern became much more visible than identity alone: people were constantly participating.

Visitors were pressing buttons repeatedly, replaying interactions, laughing together, filming each other, and physically engaging with installations in ways that passive displays did not encourage.

At that point, participation became the strongest direction for the project, eventually leading me towards tactile interaction and visitor engagement.

Building Secondary Research

Connecting secondary Research to My Observations

To support my observations, I began collecting secondary research related to:

- interactive museum engagement
- participatory exhibition design
- embodied interaction
- visitor behaviour within exhibition spaces
- button based interaction systems

I specifically focused on sources that explored how interactive artefacts influence:

- curiosity
- engagement
- dwell time
- emotional participation
- exploration
- visitor control

This research helped me understand that interactive systems are not only technological features, but behavioural triggers that shape how visitors move, respond, and participate within a space.

O'Leary (2021) explains that secondary research helps researchers understand what is already known while identifying gaps and positioning their own enquiry within wider debates.

This became important in helping me narrow an initially broad topic into a more focused investigation.

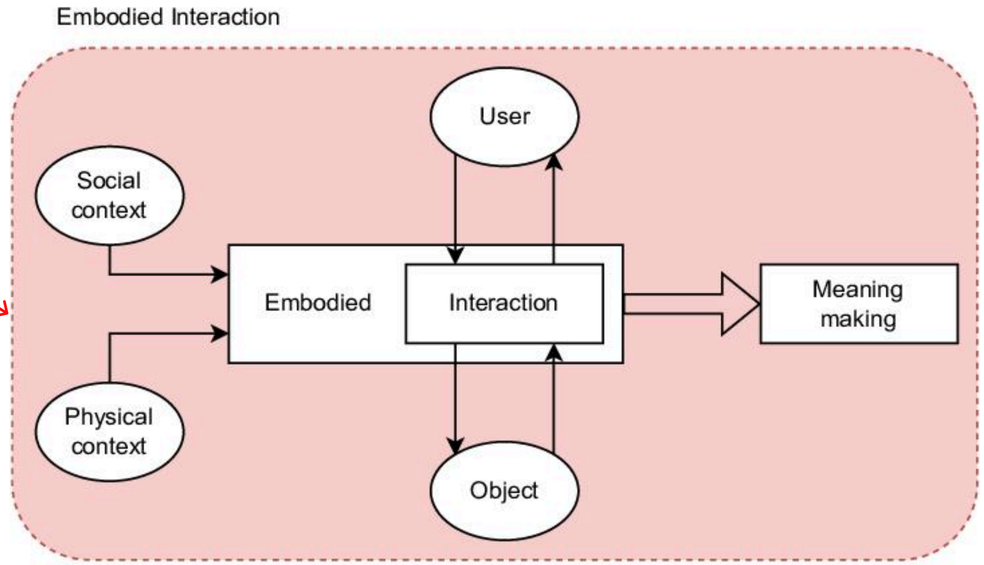


image from researchgate

Interactive Participation in Museums

Participation Through Interaction

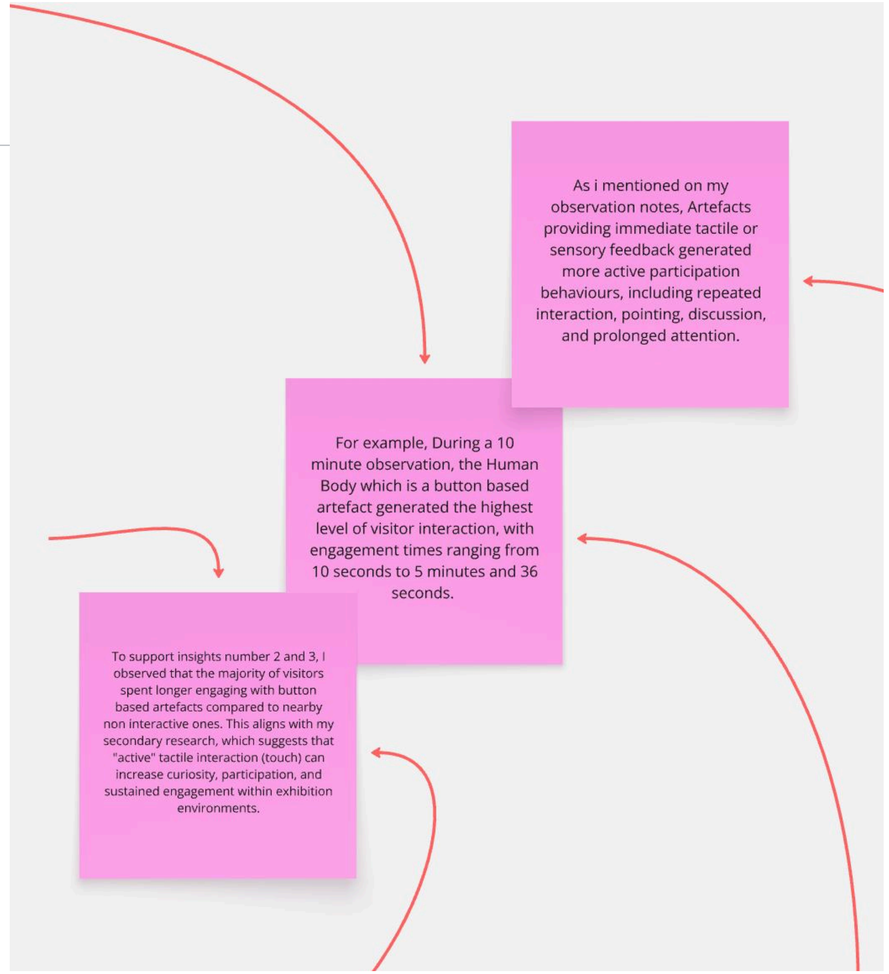
As my research developed, I became increasingly interested in how interactive exhibition design encourages visitors to move from passive observation into active participation.

Many of the sources I explored discussed how tactile interaction creates stronger emotional and behavioural engagement because visitors are physically involved in the experience rather than simply viewing it.

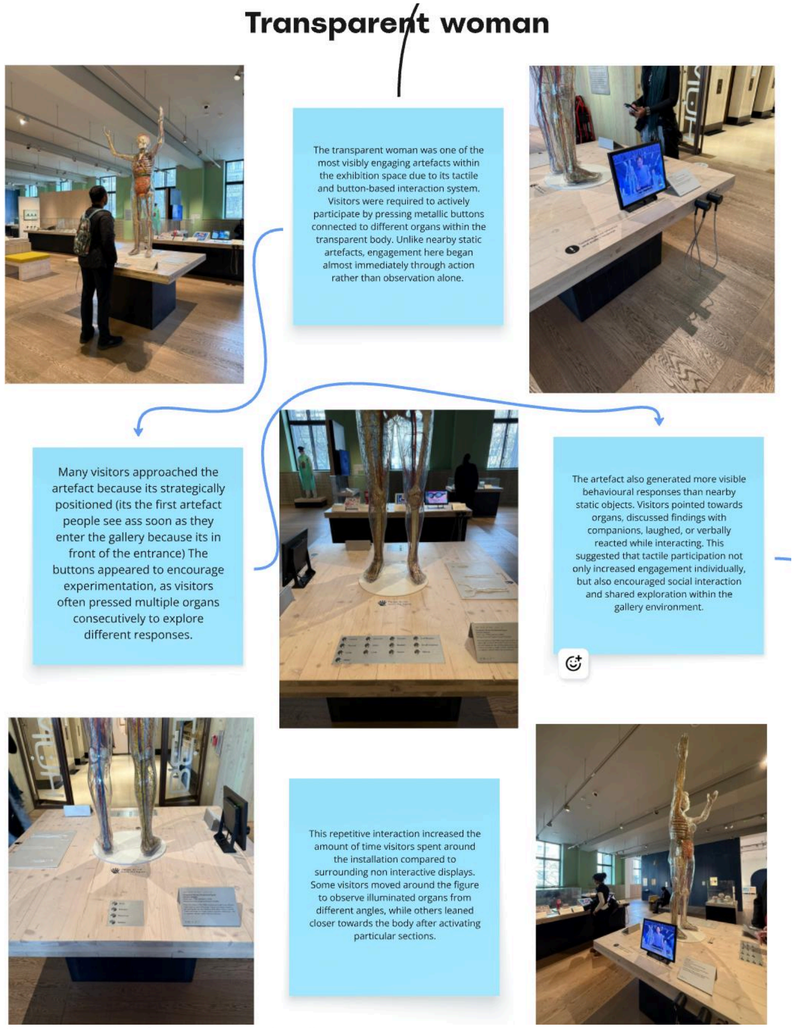
This became especially relevant to my observations around button based artefacts at the Wellcome Collection, where visitors repeatedly interacted with installations instead of engaging only once.

The research also suggested that interaction creates a stronger sense of personal agency and curiosity. Visitors are more likely to experiment, replay actions, and remain engaged for longer periods when they feel in control of the experience.

IDEO's Field Guide to Human Centred Design (2015) explains that understanding people requires experiencing situations from their perspective. This directly influenced my autoethnographic approach and encouraged me to experience the exhibition personally while observing how interaction shaped behaviour.



Screenshot taken from my research wall



Screenshot taken from my research wall

Why Button Based Artefacts Became Important

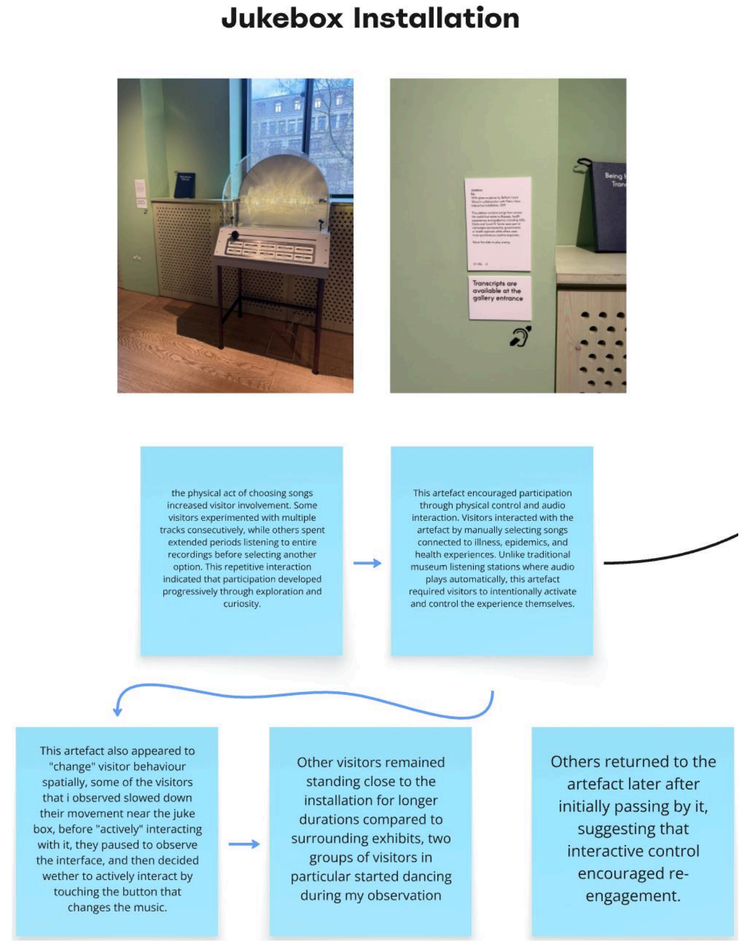
While analysing both the literature and my own observations, I realised that button based artefacts consistently produced the strongest visible behavioural responses within the exhibition. Unlike passive displays, these artefacts encouraged:

- repeated interaction
- experimentation
- shared participation
- physical engagement
- replay behaviour
- longer dwell times

The Invisible Woman installation became particularly important because visitors rarely pressed the button only once. Many visitors replayed the interaction several times, stayed longer within the area, or invited other people to participate.

Similarly, the Jukebox installation encouraged collective participation through sound, movement, and tactile control. Visitors often selected multiple tracks, danced together, or observed others before joining in themselves.

These repeated patterns suggested that tactile interaction was influencing not only engagement levels, but also social behaviour within the exhibition space.



Screenshot taken from my research wall

Research Wall and Synthesising Information

As the amount of information increased, the research wall became an important way of organising observations, references, and emerging themes visually.

Rather than simply collecting information, I started identifying recurring behavioural patterns across both secondary and primary research.

Thomas (2020) explains that analysis helps identify what happened, while synthesis helps explain what those observations mean. This became important when transforming raw observations into broader insights about participation and engagement.

Youngblood et al. (2020) also describe synthesis as a process of transforming information into insight by revealing what is happening, for whom, and why it matters.

Screenshot taken from my research wall

Linking Secondary Research to My Research Question

By combining secondary research with repeated observations at the Wellcome Collection, I became increasingly focused on one recurring behaviour:
visitors were consistently engaging more deeply with tactile, button based artefacts than with passive exhibitions.

The research helped support many of the behaviours I was observing directly within the space:

- experimentation
- replay interaction
- prolonged engagement
- social participation
- physical involvement
- curiosity driven exploration

This eventually led to my final research question, which is “how do tactile interactions with button based artefacts affect visitor engagement within the exhibition space?”

At this stage, the project became less about analysing exhibitions broadly, and more about understanding how physical interaction changes visitor behaviour, participation, and engagement within museum environments.

Interactive and Non Interactive Artefacts

During my observations, I began categorising the exhibition artefacts into two broad groups: interactive artefacts and non interactive artefacts.

This comparison became important because each category appeared to generate very different forms of visitor behaviour and engagement.

The interactive artefacts encouraged direct sensory participation through:

- button pressing
- tactile exploration
- physical activation
- audio interaction
- experimentation

In contrast, the non interactive artefacts relied more heavily on:

- visual observation
- reading
- interpretation
- reflection
- passive engagement
-

Although both forms created meaningful experiences, the interactive installations consistently generated more visible behavioural responses, particularly through movement, replay interaction, and social participation.

This comparison helped me understand how different forms of interaction can shape visitor behaviour within exhibition environments.

Photos taken by me

Observation Focus: Transparent Woman

Tactile Interaction Through Physical Control

The Transparent Woman became one of the strongest examples of tactile participation within the exhibition.

Unlike nearby static artefacts, the installation required visitors to physically activate buttons connected to different organs within the transparent body.Engagement therefore began through action rather than observation alone.

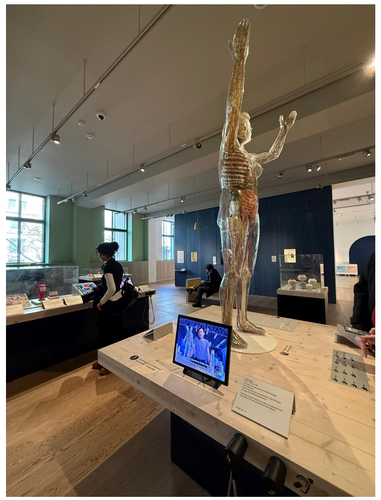
Its placement near the entrance also influenced behaviour significantl, many visitors approached the artefact almost immediately after entering the gallery, making it one of the first interactive encounters within the space.

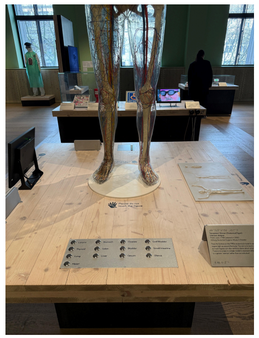
Repeated button pressing became a recurring behaviour during observation sessions. Visitors often activated multiple organs consecutively in order to compare responses and continue exploring the installation.

The artefact also generated stronger social interaction than surrounding passive artefacts.

Visitors frequently pointed towards illuminated organs, discussed discoveries with companions, or verbally reacted while interacting.

This suggested that tactile participation increased both individual curiosity and collective engagement within the exhibition environment.





Photos taken by me

Spatial Engagement Around the Transparent Woman

The Transparent Woman did not only influence interaction itself, but also the way visitors physically positioned themselves around the installation.

During several observations, visitors:

- leaned closer towards the body after activating buttons
- moved around the figure to observe illuminated organs from different angles
- paused longer compared to nearby artefacts
- replayed interactions before moving elsewhere

This repetitive exploration increased dwell time around the installation (average dwell time, from 10 seconds of interaction to over 5 minutes) and transformed the surrounding area into an active behavioural space rather than a simple viewing point.

Compared to nearby non interactive artefacts, visitors appeared more physically involved and more willing to experiment through touch and movement.

These behaviours became increasingly important in shaping my understanding of tactile engagement within exhibitions.

Observation Focus: Jukebox Installation

Participation Through Audio and Button Based Control

The Jukebox Installation encouraged participation through both tactile control and audio interaction.

Unlike traditional listening stations where sound plays automatically, visitors were required to intentionally activate the experience themselves by selecting songs manually through button based controls.

This immediately changed the relationship between visitor and artefact. Participation became active rather than passive.
Many visitors experimented by selecting multiple tracks consecutively, while others spent extended periods listening before returning to interact again.

One of the most noticeable observations was the way the installation appeared to influence movement and behaviour spatially. Some visitors slowed down near the installation before deciding whether to interact, often pausing first to observe the interface or watch others participate.

The artefact therefore created anticipation and curiosity even before direct interaction occurred.

Photos and sketches done by me

Shared Participation and Re Engagement

The Jukebox installation generated some of the strongest social behaviours observed during the project.

Visitors frequently remained near the installation for longer periods compared to surrounding exhibits, often observing other people interact before participating themselves.

During one observation session, two groups of visitors began dancing together near the installation while experimenting with different tracks.

This demonstrated how tactile and audio based interaction could transform an artefact into a temporary social environment rather than an isolated individual experience.

Another recurring behaviour was re engagement. Several visitors initially passed by the installation before returning later after hearing music or observing interaction happening nearby.

This suggested that tactile control and visible participation encouraged curiosity and prolonged engagement over time.

Other artefacts

5318008: Tactile and Sensory Interaction

The artefact 5318008 created a very different type of tactile interaction compared to the button based installations.

Unlike traditional museum installations that discourage touch, this artefact explicitly invited visitors to physically interact through the phrase “Please touch”.

This immediately disrupted conventional museum expectations and encouraged a more open and experimental atmosphere around the object.

Many visitors physically leaned closer to smell or touch the surface after noticing the invitation to interact. During one observation, a group of visitors collectively discussed the sign before deciding together to engage with the artefact.

This showed how sensory participation could influence behaviour beyond observation alone.

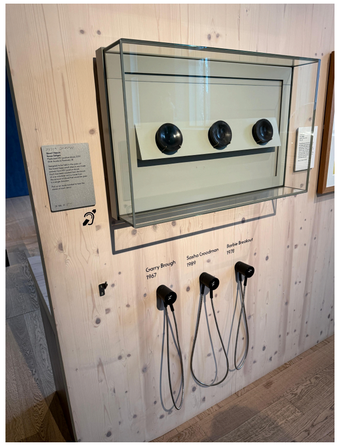
The combination of:

- touch
- smell
- physical proximity
- embodied interaction

and created a more intimate and immersive form of engagement than purely visual artefacts.

Although not button based, this artefact remained important because it reinforced the broader relationship between tactile interaction and visitor participation.

Photos taken by me



Audio Listening and visual (displays) Installations

Several artefacts within the exhibition incorporated headphone based listening experiences, including Blood Objects and the Friendship Bench installation.

Although these artefacts involved less physical interaction than the button based installations, they still encouraged intentional forms of participation through concentrated listening and visual immersion.

Many visitors paused for extended periods while using headphones, often remaining stationary and fully focused on recorded testimonies or audio narratives.

The Friendship Bench particularly encouraged prolonged engagement because of its seating arrangement, creating a quieter and more reflective experience within the gallery.

However, compared to the button based artefacts, these installations generally generated:

- fewer spontaneous reactions
- less visible experimentation
- reduced social interaction
- lower replay behaviour

Visitors often observed others before deciding whether to participate, suggesting that quieter interactions attracted attention more slowly than tactile activation through buttons and physical control.

Emerging Behavioural Difference

Passive Observation vs...

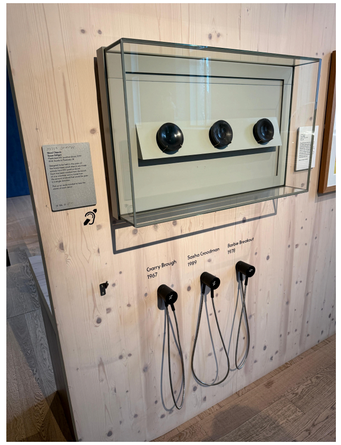
Across all observations, one of the clearest behavioural differences emerged between passive observation and active tactile participation.

Non interactive artefacts often encouraged:

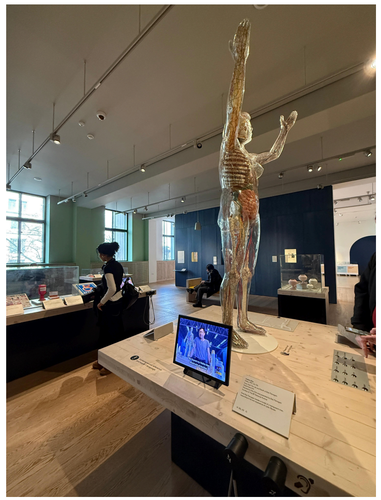
- contemplation
- slower observation
- reading and interpretation
- emotional reflection

Meanwhile the tactile and button based installations consistently generated:

- immediate behavioural responses
- experimentation
- replay interaction
- curiosity driven exploration
- social engagement
- prolonged dwell time



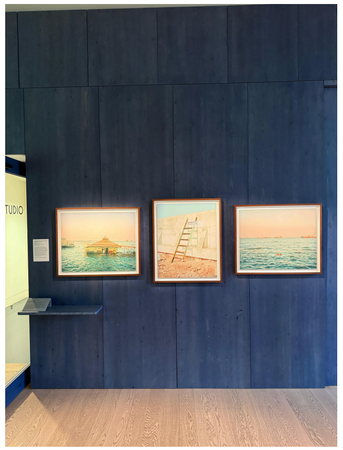
Photos taken by me



...Active Participation

Although the exhibition as a whole explored participation in multiple ways, the button based artefacts repeatedly produced the strongest visible forms of active engagement.

At this stage of the project, tactile interaction was no longer appearing as an isolated behaviour, but as a recurring pattern influencing how visitors explored and participated within the exhibition space.



Understanding Analysis and Synthesis

After completing my field observations within the Being Human exhibition, I began analysing recurring visitor behaviours connected to tactile interaction and participation.

At this stage, the project shifted from collecting observations towards identifying patterns, relationships, and behavioural themes linked to visitor engagement.

Through coding, mapping, synthesis, and comparative observation, I started understanding how button based artefacts appeared to influence curiosity, movement, participation, and prolonged engagement within the exhibition environment.

During this stage of the project, I used both analysis and synthesis to interpret the research findings.

Analysis helped me break down observations into smaller behavioural components such as:

- repeated button pressing
- selective listening
- prolonged dwell time
- attention shifting

Synthesis helped me connect these behaviours together in order to understand what they revealed about visitor engagement.

Rather than only describing what visitors were doing, I started identifying why tactile interaction appeared to create stronger participation compared to nearby non interactive artefacts.

Coding the Research; Coding Behavioural Observations

To organise my findings, I began coding recurring visitor behaviours observed throughout the exhibition.

The codes emerged directly from repeated interactions surrounding button based artefacts such as the Invisible Woman and the Jukebox installation.

Coding allowed me to move beyond isolated observations and identify behavioural themes connected to tactile engagement and participation.

Final Codes

These codes helped reveal behavioural similarities across different interactive artefacts within the exhibition space.

behavioural codes:

- Button interaction
- Replay behaviour
- Curiosity driven interaction
- Choice based interaction
- Nonlinear engagement
- Selective listening
- Attention shifting

Other codes (engagement)

- Emotional engagement
- Touch interaction
- Group participation
- Multisensory interaction
- Interpretation freedom

Journey Map

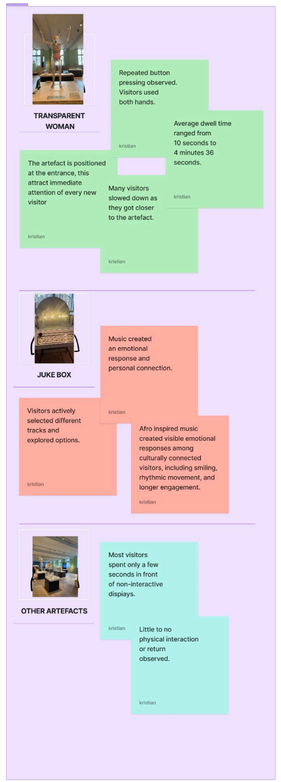
Visitor Interaction Journey

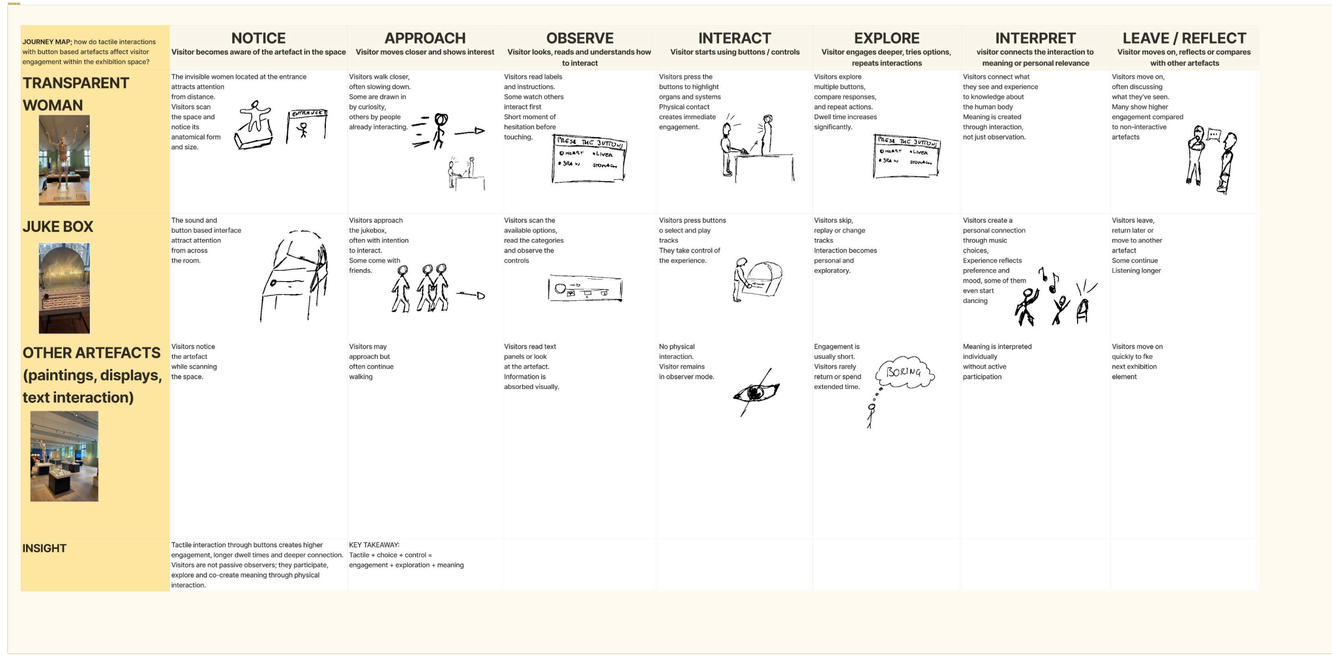
The journey map helped visualise how visitors interacted with tactile artefacts throughout different stages of engagement.

Across both the Invisible Woman and the Jukebox installation, visitors generally moved through similar stages:
NOTICE → APPROACH → OBSERVE → INTERACT → EXPLORE → INTERPRET → REFLECT

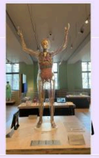
Unlike passive artefacts, engagement often continued after the first interaction.

Visitors repeatedly explored options, replayed interactions, and experimented with different outcomes through tactile participation. This suggested that engagement became progressively deeper once visitors gained physical control over the interaction.





Journey Map



TRANSPARENT WOMAN

Repeated button pressing observed. Visitors used both hands.

kristian

Average dwell time ranged from 10 seconds to 4 minutes 36 seconds.

kristian

The artefact is positioned at the entrance, this attract immediate attention of every new visitor

kristian

Many visitors slowed down as they got closer to the artefact.

kristian

JUKE BOX

Music created an emotional response and personal connection.

kristian

Visitors actively selected different tracks and explored options.

kristian

Afro inspired music created visible emotional responses among culturally connected visitors, including smiling, rhythmic movement, and longer engagement.

kristian

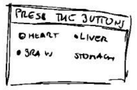
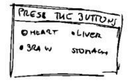
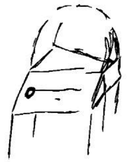
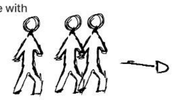
OTHER ARTEFACTS

Most visitors spent only a few seconds in front of non-interactive dispiays.

kristian

Little to no physical interaction or return observed.

kristian

JOURNEY MAP; how do tactile interactions with button based artefacts affect visitor engagement within the exhibition space?	NOTICE Visitor becomes aware of the artefact in the space	APPROACH Visitor moves closer and shows interest	OBSERVE Visitor looks, reads and understands how to interact	INTERACT Visitor starts using buttons / controls	EXPLORE Visitor engages deeper, tries options, repeats interactions	INTERPRET visitor connects the interaction to meaning or personal relevance	LEAVE / REFLECT Visitor moves on, reflects or compares with other artefacts
TRANSPARENT WOMAN 	The invisible women located at the entrance attracts attention from distance. Visitors scan the space and notice its anatomical form and size. 	Visitors walk closer, often slowing down. Some are drawn in by curiosity, others by people already interacting. 	Visitors read labels and instructions. Some watch others interact first Short moment of hesitation before touching. 	Visitors press the buttons to highlight organs and systems Physical contact creates immediate engagement. 	Visitors explore multiple buttons, compare responses, and repeat actions. Dwell time increases significantly. 	Visitors connect what they see and experience to knowledge about the human body Meaning is created through interaction, not just observation.	Visitors move on, often discussing what they've seen. Many show higher engagement compared to non-interactive artefacts
JUKE BOX 	The sound and button based interface attract attention from across the room. 	Visitors approach the jukebox, often with intention to interact. Some come with friends. 	Visitors scan the available options, read the categories and observe the controls 	Visitors press buttons o select and play tracks They take control of the experience. 	Visitors skip, replay or change tracks Interaction becomes personal and exploratory.	Visitors create a personal connection through music choices, Experience reflects preference and mood, some of them even start dancing 	Visitors leave, return later or move to another artefact Some continue Listening longer
OTHER ARTEFACTS (paintings, displays, text interaction) 	Visitors notice the artefact while scanning the space.	Visitors may approach but often continue walking	Visitors read text panels or look at the artefact. Information is absorbed visually.	No physical interaction. Visitor remains in observer mode. 	Engagement is usually short. Visitors rarely return or spend extended time. 	Meaning is interpreted individually without active participation	Visitors move on quickly to fke next exhibition element
INSIGHT	Tactile interaction through buttons creates higher engagement, longer dwell times and deeper connection. Visitors are not passive observers; they participate, explore and co-create meaning through physical interaction.	KEY TAKEAWAY: Tactile + choice + control = engagement + exploration + meaning					

Spatial Relationship Diagram

Spatial Behaviour and Movement Patterns

The spatial relationship diagram helped identify how interactive artefacts influenced visitor movement throughout the exhibition environment.

The Invisible Woman became one of the strongest attraction points due to:

- its position near the entrance
- immediate visibility
- tactile button based interaction

Similarly, the Jukebox installation attracted attention through sound, visible participation, and physical interaction.

The observations also revealed that visitors were often influenced by watching other people interact with the artefacts before participating themselves.

This created moments of shared curiosity and social participation within the exhibition space.

SPATIAL RELATIONSHIP DIAGRAM

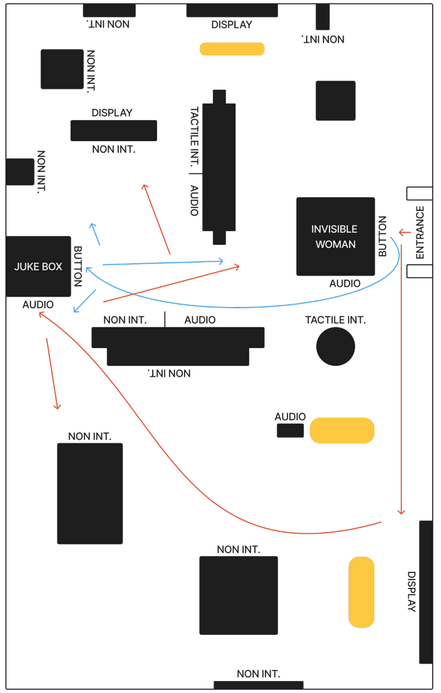
This diagram maps the position of interactive and passive artefacts and the spatial relationships that shape visitor movement, attention and engagement within the exhibition

Immediate visual attraction
The invisible woman is directly in front of the entrance making it the first point of attraction for almost all visitors

Secondary attraction
The Juke box, an other button based audio artefact, attracts visitors from across the room, many visitors hear or see other engage with it and are curious to try it, thats how i discovered the juke box too, after interacting with the invisible women, i noticed a group of people dancing and changing music on the juke box and i went straight to it

ADDITIONAL OBSERVATIONS FROM FIELD RESEARCH

- People often return to the same interactive exhibits
- Visitors Spend more time on the interactive artefacts when tey are less crowded
- Groups tend to split anc reunite at high attraction points, particularly button based like invisible woman or juke box
- Children are strongly attracted to the invisible woman
- The interactive artefacts make visitor repeat some actions, like pressing the same buttons for example



Visitor behaviour

- Visitors don't follow a fixed path, movement is organic, fluid and influenced by curiosity, sound, visibility and other visitors' behaviour.
- Most visitors start at the Invisible Woman due to its direct visibility from the entrance.
- After interacting with the invisible woman, visitors either move left towards the large TV on the wall and continue exploring the perimeter, or they are attracted across the room to the Jukebox by sound and social triggers
- Many visitors stop to watch others laugh, record videos or dance at the Jukebox before engaging
- Visitors often revisit artefacts (especially Invisible Woman and Jukebox) multiple times.
- Seating is used for resting or observing displays

INSIGHTS

- Focal point: The Invisible Woman is the strongest focus point due to its position in front of the entrance, its the first thing a visitor look at when he enter the room
- Audio and tactile (button based) attraction: The Jukebox has a strong pull factor through music and interaction sounds, and by letting people control the music, it attracts visitors from long distances
- Random pathways: Many visitors follow random pathways that end up in a loop, encountering displays and and static artefacts along the way, only to end up interacting again with invisible woman or juke box
- SOCIAL INFLUENCE: laughs, movement, and "replay" behaviour from others heavily influence visitor decisions and movement

GUIDE

BLACK, ARTEFACTS:
- Display (tv, short documentary)
- Button (button based artefact)
- Tactile int (non button based, but you can interact by touching the artefact)
- audio (audio based / has also audio content)

YELLOW: couches, seats

Movement patterns observed (arrows guide)

Direct attraction to the juke box
Some visitors go straight from the invisible women to the juke box after seeing others interact , hearing music or dancing

Perimeter exploration
Visitors start at the invisible women, then move on the left along the perimeter towards the biggest screen on the right lower corner of the room, and from there, eventually, discover the juke box

? Random exploration
most of the visitors that i observed took random paths

Passive engagement
Visitors often pause to watch others interact, especially at the Invisible Woman and Jukebox, creating moments of shared entertainment and social connection

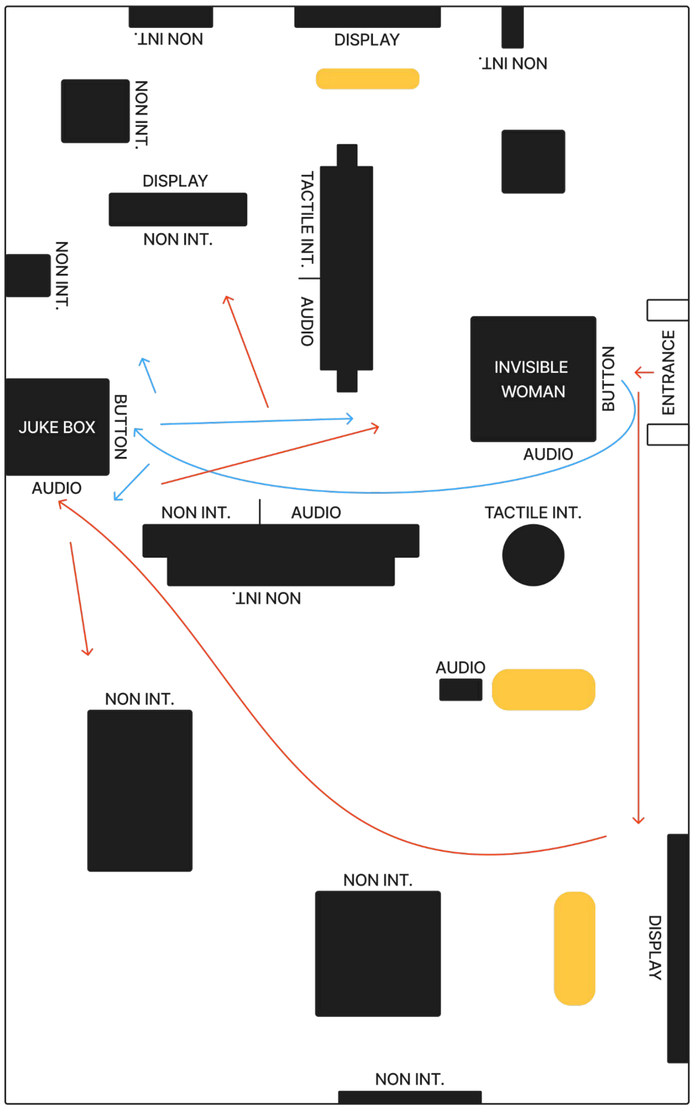
Displays
Most of the screens ar positioned along ine walls, allowing content to be visible from multiple points in the room without creating congestion in the central area, the only monitor that is not positioned on a wall, is in the left corner of the room, which makes it still visible from different positions and angles.

SPATIAL RELATIONSHIP DIAGRAM

This diagram maps the position of interactive and passive artefacts and the spatial relationships that shape visitor movement, attention and engagement within the exhibition

Immediate visual attraction
The invisible woman is directly in front of the entrance making it the first point of attraction for almost all visitors

Secondary attraction
The Juke box, an other button based audio artefact, attracts visitors from across the room, many visitors hear or see other engage with it and are curious to try it, thats how i discovered the juke box too, after interacting with the invisible women, i noticed a group of people dancing and changing music on the juke box and i went straight to it



GUIDE

BLACK, ARTEFACTS:
- Display (tv, short documentary)
- Button (button based artefact)
- Tactile int (non button based, but you can interact by touching the artefact)
- audio (audio based / has also audio content)

YELLOW: couches, seats

Movement patterns observed (arrows guide)

Direct attraction to the juke box
Some visitors go straight from the invisible women to the juke box after seeing others interact , hearing music or dancing

Perimeter exploration
Visitors start at the invisible women, then move on the left along the perimeter towards the biggest screen on the right lower corner of the room, and from there, eventually, discover the juke box

? Random exploration
most of the visitors that i observed took random paths

ADDITIONAL OBSERVATIONS FROM FIELD RESEARCH
- People often return to the same interactive exhibits
- Visitors Spend more time on the interactive artefacts when tey are less crowded
- Groups tend to split anc reunite at high attraction points, particularly button based like invisible woman or juke box
- Children are strongly attracted to the invisible woman
- The interactive artefacts make visitor repeat some actions, like pressing the same buttons for example

Passive engagement
Visitors often pause to watch others interact, especially at the Invisible Woman and Jukebox, creating moments of shared entertainment and social connection

Visitor behaviour
- Visitors don't follow a fixed path, movement is organic, fluid and influenced by curiosity, sound, visibility and other visitors' behaviour.
- Most visitors start at the Invisible Woman due to its direct visibility from the entrance.
- After interacting with the invisible woman, visitors either move left towards the large TV on the wall and continue exploring the perimeter, or they are attracted across the room to the Jukebox by sound and social triggers
- Many visitors stop to watch others laugh, record videos or dance at the Jukebox before engaging
- Visitors often revisit artefacts (especially Invisible Woman and Jukebox) multiple times.
- Seating is used for resting or observing displays

Displays
Most of the screens ar positioned along ine walls, allowing content to be visible from multiple points in the room without creating congestion in the central area, the only monitor that is not positioned on a wall, is in the left corner of the room, which makes it still visible from different positions and angles.

INSIGHTS
- Focal point: The Invisible Woman is the strongest focus point due to its position in front of the entrance, its the first thing a visitor look at when he enter the room
- Audio and tactile (button based) attraction: The Jukebox has a strong pull factor through music and interaction sounds, and by letting people control the music, it attracts visitors from long distances
- Random pathways: Many visitors follow random pathways that end up in a loop, encountering displays and and static artefacts along the way, only to end up interacting again with invisible woman or juke box
- SOCIAL INFLUENCE: laughs, movement, and "replay" behaviour from others heavily influence visitor decisions and movement

Spatial Relationship Diagram

Spatial Relationship Diagram

As observations accumulated, recurring patterns began to appear across multiple artefacts.

Visitors rarely interacted with exhibitions in a fully linear or passive way, instead, interaction became:

- selective
- exploratory
- self directed

This behaviour became especially visible around button based artefacts, where visitors actively controlled the pace and direction of interaction themselves.

Compared to nearby passive artefacts, tactile artefacts consistently generated:

- longer dwell times
- repeated participation
- physical exploration
- stronger social interaction

Pattern Clusters

Self directed participation

Visitors often selected specific interactions based on personal curiosity rather than following exhibitions in a fixed sequence.

This behaviour appeared strongly connected to button based artefacts where visitors could actively choose what to interact with.

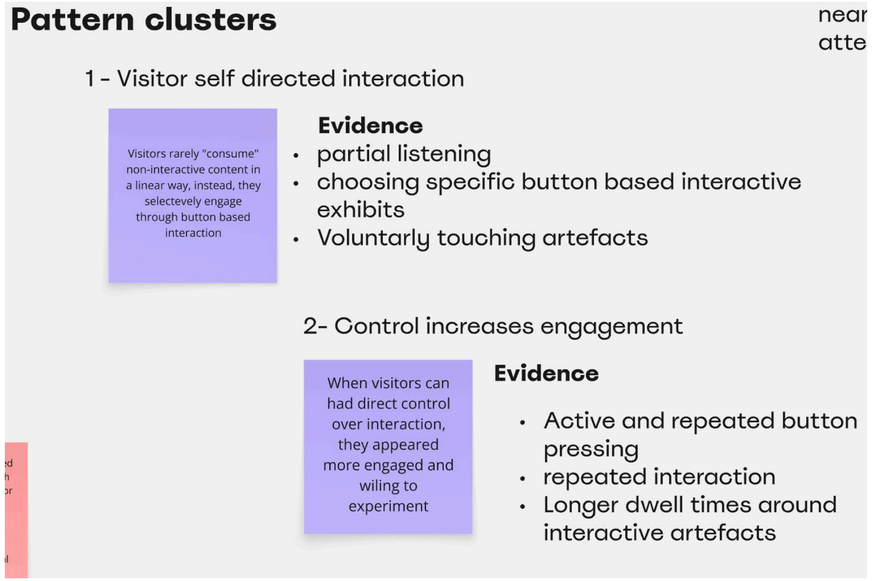
Control increases engagement

When visitors gained direct physical control over interaction, engagement appeared to increase significantly.

Visitors frequently:

- replayed interactions
- experimented with different buttons
- explored multiple outcomes
- remained longer within the installation space

This behaviour was less visible around non interactive artefacts.



Screenshot taken from my research wall

Critical Tension

While analysing the observations, I identified a tension between how museums structure experiences and how visitors actually engage with them.

Many exhibitions are designed around curated narratives and intended sequences of interaction.

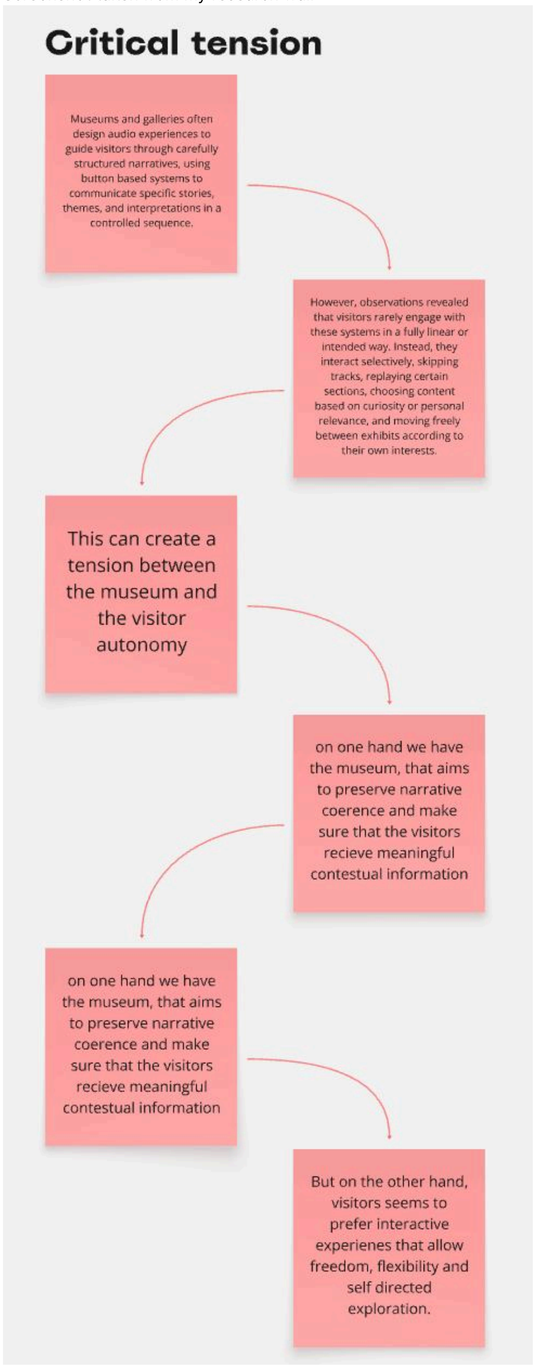
However, the observations suggested that visitors often interacted selectively rather than linearly.

Instead of following fixed paths, visitors:

- skipped sections
- replayed interactions
- selectively listened
- moved freely between artefacts
- prioritised curiosity and personal interest

This suggested that tactile interaction may encourage more autonomous and self directed engagement within exhibition environments.

Screenshot taken from my research wall



Supporting Observational Evidence

Several repeated observations supported the emerging behavioural patterns, for example:

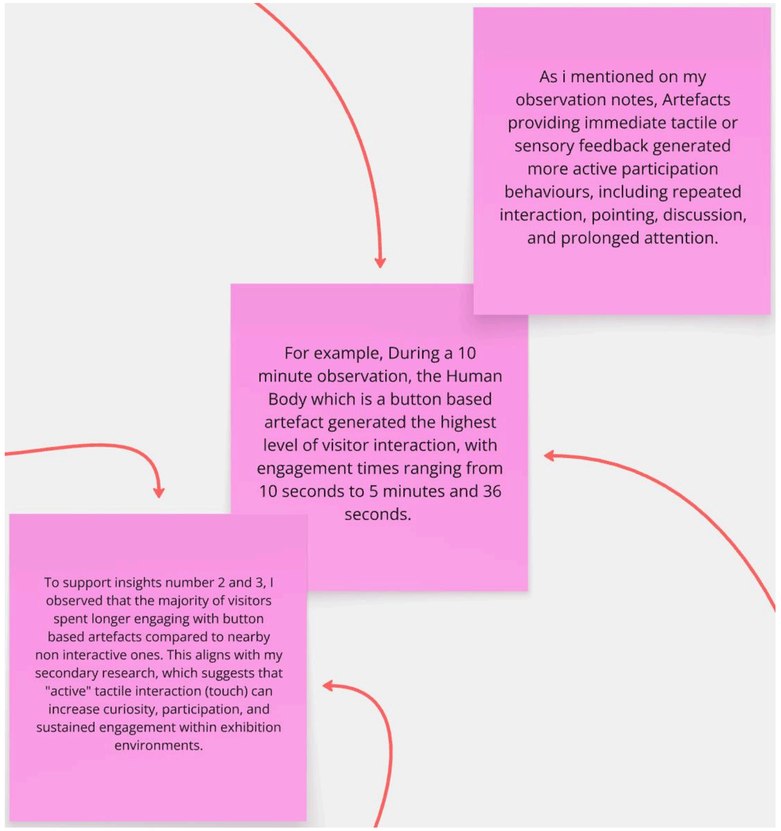
- visitors repeatedly pressed buttons to explore different outcomes
- groups gathered around tactile artefacts
- visitors spent longer engaging with interactive installations
- interaction generated discussion, pointing, and shared reactions
- some visitors returned to the same artefacts multiple times

During one observation session, the Invisible Woman installation generated engagement times ranging from a few seconds to over five minutes, significantly longer than nearby passive artefacts.

These repeated behaviours strengthened the relationship between tactile interaction and prolonged engagement.

Synthesised Insights

Insight 1
Visitors consistently spent more time engaging with button based artefacts than nearby non interactive displays.
Insight 2
Physical control and tactile interaction appeared to increase curiosity, experimentation, and repeated participation.
Insight 3
Interactive artefacts encouraged visitors to become active participants rather than passive observers.
Insight 4
Social interaction frequently emerged around tactile installations, particularly when visitors observed others participating first.



Screenshot taken from my research wall

Insight Statement

The analysis and synthesis process revealed that tactile button based artefacts generated stronger visitor engagement compared to nearby passive displays.

Artefacts such as the Invisible Woman and the Jukebox encouraged:

- repeated interaction
- curiosity driven exploration
- prolonged dwell time
- physical participation
- social engagement

These findings suggest that tactile interaction can transform visitors from passive observers into active participants within exhibition environments.

Insight Statement

Interactive, button based artefacts like the invisible woman or the juke box suggested that visitors engaged more actively with button based artefacts than the non interactive ones.
Button based artefacts generated longer dwell times, repeated interaction and more exploratory behaviour through tactile participation.
These findings makes me think that physical interaction can transform visiotrs from passive observers into active participants.

Screenshot taken from my research wall

Opportunity Framing and Design Direction

From Insights to Opportunities

Translating Research into Design Opportunities

After analysing the behavioural patterns observed throughout the exhibition, I began exploring how these findings could inform possible interaction opportunities within museum environments.

The research repeatedly suggested that tactile button based interaction encouraged:

- curiosity driven exploration
- prolonged engagement
- replay behaviour
- active participation
- visitor autonomy

At this stage of the project, the focus shifted from understanding behaviour towards identifying how these interaction patterns could inspire future exhibition experiences.

Rather than redesigning the entire exhibition, I became interested in how tactile interaction itself could support more meaningful and participatory forms of engagement.

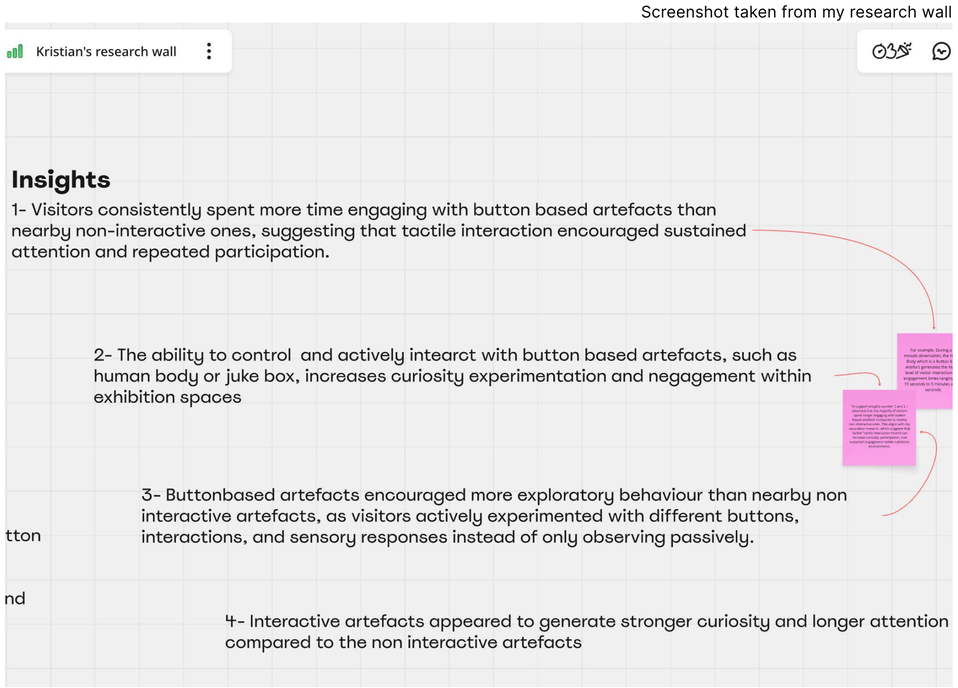
Emerging Opportunity Areas

Several recurring opportunities emerged from both the primary and secondary research, one of the strongest opportunities involved creating interaction systems that allow visitors to actively control their experience through tactile participation rather than passive observation alone.

The observations also suggested that visitors respond strongly to experiences that encourage:

- experimentation
- replay interaction
- multisensory engagement
- visible participation from others

Screenshot taken from my research wall



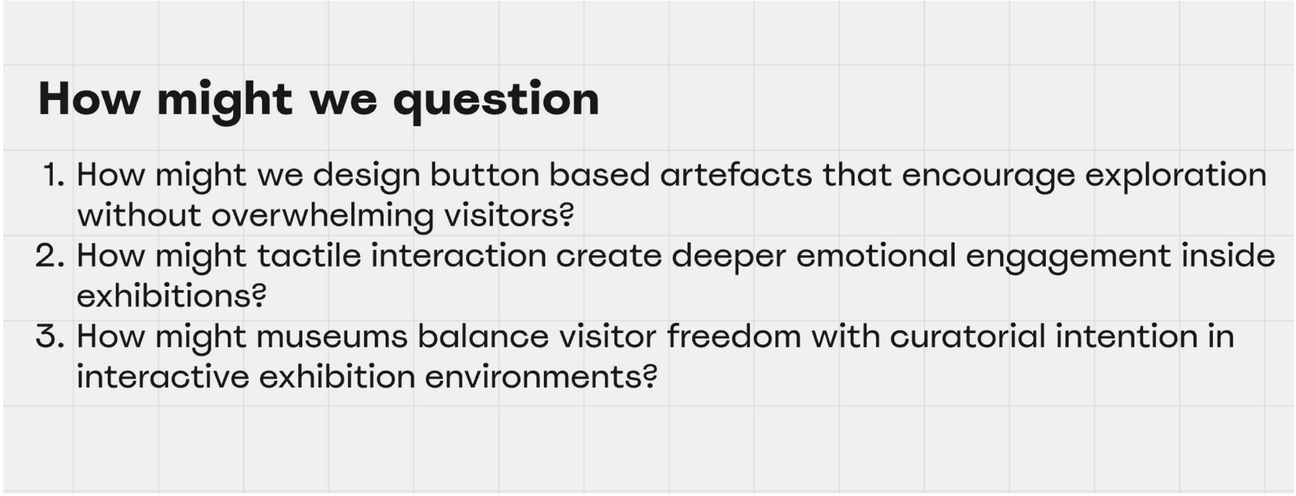
How Might We Questions (refined ones)

1.
How might tactile button based interaction encourage deeper curiosity and exploration within exhibition environments?

2.
How might visitors become more active participants rather than passive observers within museum spaces?

3.
How might interactive artefacts balance visitor freedom with meaningful engagement and interpretation?

4.
How might tactile interaction encourage replay behaviour and prolonged engagement without overwhelming visitors?



Screenshot taken from my research wall (unrefined how might we questions)

Connecting Research to the Final Artefact

The opportunity framing stage helped clarify which aspects of the research were most important to carry forward into the final artefact.

The final concept became centred around:

- tactile participation
- physical interaction
- visitor autonomy
- curiosity driven exploration
- replay engagement
- active rather than passive participation

Screenshot taken from my research wall

Introducing the Final Artefact

System Map

I wanted to create a final artefact that could visually combine the different observations, behaviours, interaction patterns, and insights identified throughout the research process.

I decided to produce a system map that explores how visitors move through the exhibition space and interact with different artefacts.

The map focuses particularly on tactile button based artefacts such as the Invisible Woman and the Jukebox installation, while also comparing them to nearby non interactive artefacts observed within the Being Human exhibition.

The purpose of the artefact was to transform fragmented observations into a clearer visual understanding of how tactile interaction influences visitor engagement, participation, movement, and curiosity within exhibition environments.

Why I Chose a System Map

I chose to create a system map because many of the behaviours I observed were interconnected rather than isolated.

Throughout the observations, visitor engagement appeared to be influenced by multiple factors simultaneously, including:

- tactile interaction
- physical movement
- audio feedback
- curiosity
- replay behaviour
- social interaction
- visibility within the space

Instead of presenting these behaviours separately, the system map allowed me to visualise how they relate to one another across the entire exhibition environment.

This approach also helped demonstrate that visitor engagement is not produced by a single artefact alone, but by a wider interaction system involving space, behaviour, curiosity, and participation.

Mapping Visitor Behaviour

One of the main focuses of the system map was understanding how visitors moved through the exhibition and how tactile interaction influenced those movements.

The observations suggested that visitors rarely followed a fixed or linear pathway. Instead, movement appeared fluid and curiosity driven.

Many visitors:

- paused after noticing visible interaction from others
- approached artefacts after hearing sound or music
- returned to previously visited installations
- explored selectively rather than sequentially
- replayed interactions multiple times

The map therefore visualises the exhibition as a dynamic behavioural system rather than a static viewing experience.

image from MuseumNext

Comparing Interactive and Passive Artefacts

The system map also divides the exhibition into two broad categories:

- interactive button based artefacts
- non interactive displays

This comparison became important throughout the research because the two categories appeared to generate very different forms of engagement.

Button based artefacts often encouraged:

- physical participation
- experimentation
- prolonged dwell time
- repeated interaction
- visible behavioural responses

In contrast, nearby non interactive artefacts generally encouraged quieter and shorter forms of engagement centred around observation and interpretation.

This comparison helped reinforce one of the main findings of the project: tactile interaction appeared to transform visitors from passive observers into more active participants.

image from MuseumNext

The Interaction Process

The central section of the system map focuses on what happens during tactile interaction itself.

Through repeated observations, several behavioural stages began to emerge:

1. visual attraction
2. approach and curiosity
3. physical interaction
4. replay behaviour
5. exploration and experimentation

This sequence appeared consistently around the Invisible Woman and Jukebox artefacts, where visitors often interacted repeatedly rather than engaging only once.

The observations suggested that tactile interaction creates a more active and self directed relationship between the visitor and the exhibition environment.

Behavioural Outcomes Observed

The map also identifies several behavioural effects repeatedly observed around tactile installations.

These included:

- longer dwell times
- repeated button pressing
- group interaction
- emotional reactions
- replay behaviour
- physical proximity to artefacts
- collaborative exploration

One important pattern was that participation often became socially visible. Visitors frequently observed other people interacting before deciding to engage themselves.

This created moments where interaction spread socially throughout the exhibition space rather than remaining individual.

Visitor Archetypes

While analysing the observations, I also noticed that visitors engaged with tactile artefacts in different ways.

The system map therefore introduces several behavioural archetypes, including:

- the explorer
- the selective visitor
- the social visitor
- the passive observer

These archetypes helped show that engagement was not identical for every visitor,however, tactile button based interaction consistently appeared to increase participation across multiple visitor types.

This reinforced the idea that tactile interaction can support different forms of engagement while still encouraging active participation.

Connecting the Artefact to the Research Question

The final artefact helped bring together the entire research process and directly respond to the research question: How do tactile interactions with button based artefacts affect visitor engagement within the exhibition space?

The observations, coding, behavioural analysis, and system mapping collectively suggested that tactile button based interaction increases:

- curiosity
- physical participation
- replay behaviour
- prolonged engagement
- visitor autonomy
- social interaction

The project therefore indicates that tactile interaction plays an important role in transforming museum visitors from passive observers into active participants within exhibition environments.

SYSTEM MAP

This map explores how visitors move through the Wellcome Collection and interact with artefacts, in particular tactile button based such as Invisible Woman and the Jukebox. It focuses on how physical interaction changes visitor behaviour, attention, participation, and movement within the exhibition environment.

Research question

How do tactile interactions with button based artefacts affect visitor engagement within the exhibition space?

The interaction system (what happens)

PHYSICAL BUTTON PRESSING: Interaction requires physical action rather than passive observation.

IMMEDIATE FEEDBACK: Visitors received instant responses through multiple sensory channels -organs illuminate -music starts -audio changes -visual response is activated

REPLAY BEHAVIOUR: Visitors repeatedly press buttons to test different responses and explore content selectively.

SELF DIRECTED INTERACTION: Visitors created their own personalised pathways instead of following fixed exhibition sequences.

Entry and visitor flow

GUIDE

OUTCOMES

OBSERVATIONS

OTHER ARTEFACTS

PATTERNS/BEHAVIOUR

INSIGHTS / EVIDENCE

INTERACTIVE (BUTTON BASED) ARTEFACTS

VISITOR ENTERS EXHIBITION
Visitors initially scan the environment visually before deciding where to move or stop

VISUAL ATTRACTION
Visitors are drawn in by different environmental factors such as:
-movement
-sound
-visible interaction
-people gathering

DECISION TO APPROACH
Interactive artefacts attracted attention more quickly because they see other people interacting with it.

Influenced by:
-interest
-knowledge
-mood
-curiosity
-social context
- group dynamics

BUTTON BASED INTERACTION ARTEFACTS:
these are artefacts that require physical interaction in order to activate content visitors became active participants by pressing buttons, replaying interactions, and controlling their own experience within the exhibition.

NON INTERACTIVE ARTEFACTS:
Passive artefacts mainly generated visual observation and shorter engagement periods

observational research suggested that tactile and controllable interaction increased visitor engagement within the exhibition space. During observations, button based artefacts consistently generated longer dwell times, replay behaviour, experimentation, and collaborative participation compared to nearby non interactive and nonbutton based artefacts.

Behavioural effects (results)

LONGER DWELL TIME
Visitors spent noticeably longer around button-based artefacts compared to nearby passive displays
ENGAGEMENT TIMES:
from approximately 10 seconds to over 5 minutes.

CURIOSITY DRIVEN PARTICIPATION
Physical interaction encouraged experimentation, repeated engagement, and active exploration.

PHYSICAL ENGAGEMENT
Visitors came closer, repositioned themselves, pointed towards buttons to other people they were with, and physically interacted with the installations.

SOCIAL INTERACTION
Participation often became collective, with group of visitors encouraging each others to interact as well.

ACTIVE PARTICIPATION
Button based artefacts transformed visitors from passive observes into active participants by encouraging tactile interaction, curiosity and prolonged engagement

Visitor archetypes

THE EXPLORER
-he presses multiple buttons
-Experiments freely
-Replays interactions
-Spends more time engaging

THE SELECTIVE VISITOR
-He choses only the most popular artefacts, which he chose by observing which one has the most visitors around it
-skips certain content
-creates his own personalised pathways
-engages through personal interest

THE SOCIAL VISITOR
-He comes with a group of people, and interact in groups
-they react together in groups, and share thoughts and experiences together

THE PASSIVE OBSERVER
-Mainly watches
-engages briefly
-Moves quickly through the space

Evidence from the observation

-Interaction times ranged from 10 seconds to 5+ minutes
-Visitors often pressed multiple buttons in different order, just a few of them following the specific order suggested by the button interface
-many visitors revised a specific artefact (the invisible woman in particular is one of the most popular artefact, one visitor came back to interact with it 3 times)
-Groups frequently gathered around interactive installations (two groups of people started interacting with the juke box and dacing together"

Other forms of interaction (non button based)

Not all participation happened through button interaction. Some visitors engaged through listening, reading, smelling, observation, or emotional reflection. These interactions created quieter but still meaningful forms of engagement within the exhibition space.

Non button based artefacts usually generated shorter interactions, but some sensory and audio installations encouraged slower and more reflective engagement. Visitors often paused briefly, read information panels, listened selectively, or observed quietly before moving through the space.

Insights from the observation

-Button based interaction encourages curiosity, experimentation and dwell time.

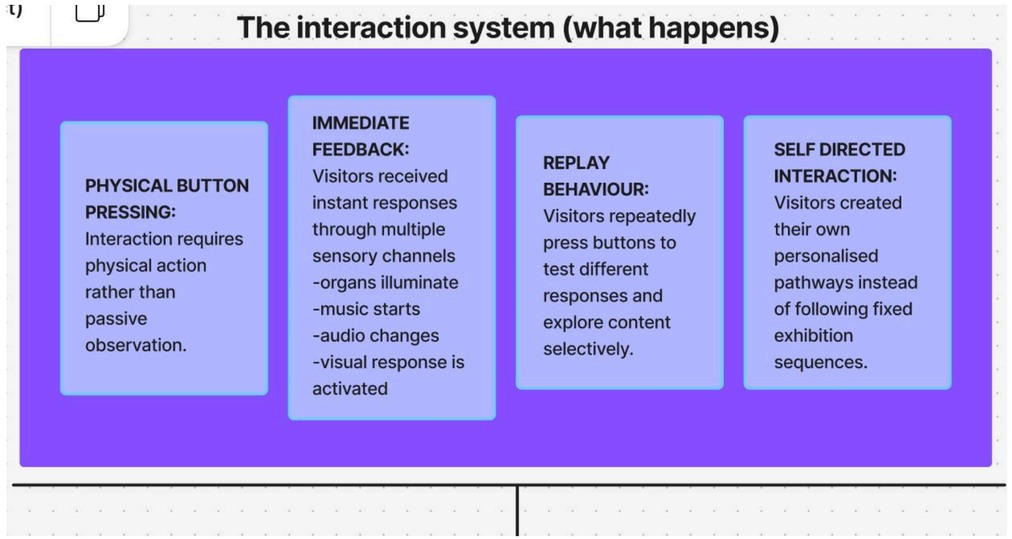
-Physical interaction makes visitors feel more involved and in control

-Visitors "replaying" their actions indicates deeper engagement and content exploration

-Social presence amplifies participation and interaction

-Tactile artefacts disrupt passive behaviour and create involvement patterns

Screenshots of Less Legible Sections from the Previous Slide



Entry and visitor flow

GUIDE

OUTCOMES

OBSERVATIONS

OTHER ARTEFACTS

PATTERNS/BEHAVIOUR

INSIGHTS / EVIDENCE

INTERACTIVE (BUTTON BASED) ARTEFACTS

VISITOR ENTERS EXHIBITION
Visitors initially scan the environment visually before deciding where to move or stop

VISUAL ATTRACTION
Visitors are drawn in by different environmental factors such as:
-movement
-sound
-visible interaction
-people gathering

DECISION TO APPROACH
Interactive artefacts attracted attention more quickly because they see other people interacting with it.

Influenced by:
-Interest
-knowledge
-mood
-curiosity
-social context
- group dynamics

BUTTON BASED INTERACTION ARTEFACTS:
these are artefacts that require physical interaction in order to activate content visitors became active participants by pressing buttons, replaying interactions, and controlling their own experience within the exhibition.

NON INTERACTIVE ARTEFACTS:
Passive artefacts mainly generated visual observation and shorter engagement periods

Evidence from the observation

-Interaction times ranged from 10 seconds to 5+ minutes
-Visitors often pressed multiple buttons in different order, just a few of them following the specific order suggested by the button interface
-many visitors revised a specific artefact (the invisible woman in particular is one of the most popular artefact, one visitor came back to interact with it 3 times)
-Groups frequently gathered around interactive installations (two groups of people started interacting with the juke box and dacing together"

Other forms of interaction (non button based)

Not all participation happened through button interaction. Some visitors engaged through listening, reading, smelling, observation, or emotional reflection. These interactions created quieter but still meaningful forms of engagement within the exhibition space.

Non button based artefacts usually generated shorter interactions, but some sensory and audio installations encouraged slower and more reflective engagement. Visitors often paused briefly, read information panels, listened selectively, or observed quietly before moving through the space.

Insights from the observation

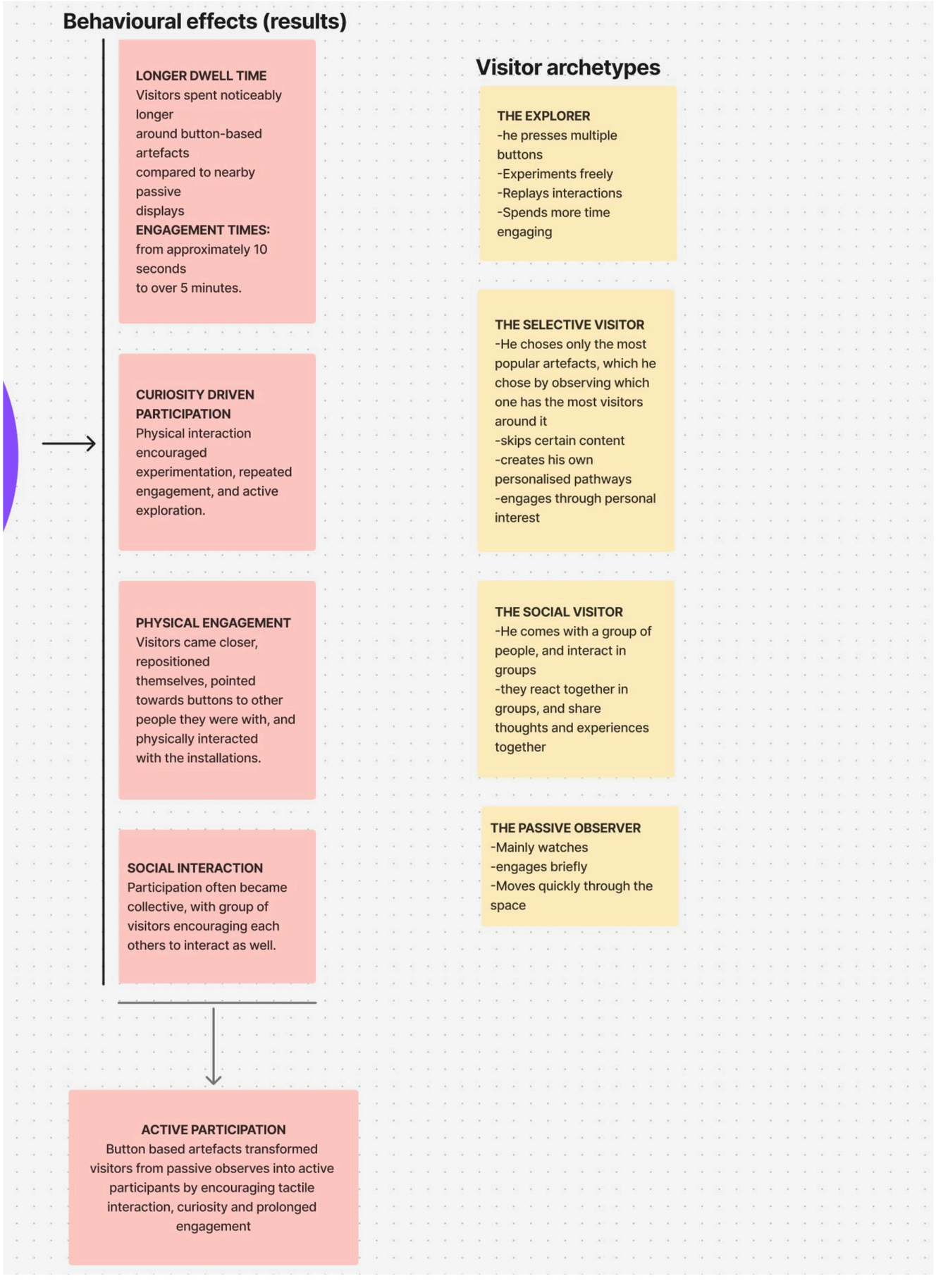
-Button based interaction encourages curiosity, experimentation and dwell time.

-Physical interaction makes visitors feel more involved and in control

-Visitors "replaying" their actions indicates deeper engagement and content exploration

-Social presence amplifies participation and interaction

-Tactile artefacts disrupt passive behaviour and create involvement patterns


Behavioural effects (results)
LONGER DWELL TIME
Visitors spent noticeably longer around button-based artefacts compared to nearby passive displays
ENGAGEMENT TIMES:
from approximately 10 seconds to over 5 minutes.
CURIOSITY DRIVEN PARTICIPATION
Physical interaction encouraged experimentation, repeated engagement, and active exploration.
PHYSICAL ENGAGEMENT
Visitors came closer, repositioned themselves, pointed towards buttons to other people they were with, and physically interacted with the installations.
SOCIAL INTERACTION
Participation often became collective, with group of visitors encouraging each others to interact as well.
ACTIVE PARTICIPATION
Button based artefacts transformed visitors from passive observes into active participants by encouraging tactile interaction, curiosity and prolonged engagement
Visitor archetypes
THE EXPLORER
-he presses multiple buttons
-Experiments freely
-Replays interactions
-Spends more time engaging
THE SELECTIVE VISITOR
-He choses only the most popular artefacts, which he chose by observing which one has the most visitors around it
-skips certain content
-creates his own personalised pathways
-engages through personal interest
THE SOCIAL VISITOR
-He comes with a group of people, and interact in groups
-they react together in groups, and share thoughts and experiences together
THE PASSIVE OBSERVER
-Mainly watches
-engages briefly
-Moves quickly through the space

Principle 03: Futures thinking

This project connects to futures thinking through its exploration of how interactive exhibitions may become increasingly self directed, immersive and participatory in the future.

rather than passively consuming information in a fixed sequence, visitors often created their own routes, selectively interacted with artefacts and replayed experiences based on curiosity and personal interest

Principle 04: Human equity and justice

This principle became relevant through the observation that visitors did not engage with exhibition content in different ways, rather than engaging with artefacts uniformly, visitors appeared to:

- selectively interact with certain instillations
- Spend different amounts of time engaging
- Respond emotionally in different ways (laughing or talking with eachother if the were in groups)
- Create personal pathways through the exhibition

For example, within the juke box installation, some visitors responded through dancing, laughter, discussion and repeated interaction, while others just observed briefly

Principle 05: Activism and advocacy

This projects connects to activism and advocacy through its focus on participation, accessibility and visitor inclusion within the exhibition environments.

My project also highlighted the importance of having exhibition environments that accommodate different behaviours, attention patterns and interaction preferences.

Reflection and Conclusion

Reflecting on the Research Process

Throughout this project, my understanding of exhibition engagement changed significantly.

At the beginning of the process, the investigation started broadly around trust and identity within museum environments. However, through repeated observation, secondary research, mapping, and synthesis, the project gradually became more focused on participation and tactile interaction.

One of the most important parts of the process was learning how research questions evolve through observation and analysis rather than remaining fixed from the beginning.

As the project developed, button based artefacts repeatedly emerged as the strongest behavioural focus within the exhibition, eventually shaping the final research direction.

What I Learned

One of the most valuable aspects of the project was learning how behavioural observation can reveal patterns that may not immediately appear through simple viewing alone.

This helped me better understand:

- how visitors physically engage with exhibitions
- how tactile interaction influences curiosity
- how participation can become socially visible
- how interaction changes movement within space
- how synthesis transforms observation into insight

Final Conclusion

This project explored how tactile button based interaction influences visitor engagement within exhibition spaces through observation, behavioural analysis, mapping, and synthesis.

Across both the primary and secondary research, interactive artefacts consistently generated stronger visible participation than nearby non artefacts.

The findings suggest that tactile interaction can transform visitors from passive observers into active participants by encouraging physical engagement and behavioural involvement within exhibition environments.

CTS Essay n.1

Helen Armstrong's text What is Participatory Design? explores how design has shifted from a process controlled entirely by designers towards something more collaborative and open. Traditionally, graphic design was often centred around control, where designers decided what audiences would see and how information would be communicated. In this model, the audience mainly acted as passive viewers receiving a finished outcome.

However, Armstrong explains that participatory design changes this relationship by involving users directly in the creation of content and meaning. Instead of delivering fully fixed products, designers create systems that allow people to interact, contribute, and shape the final experience themselves. This means users are no longer just observers, but active participants within the design process.
One of the most important ideas in the text is that participatory design relies on contribution in order to function. Armstrong argues that participation has become increasingly common because digital platforms and online tools have made creating and sharing content "cheap, easy, and fun" (Armstrong, n.d.). Social media platforms such as TikTok or YouTube demonstrate this clearly, as the content is largely created and distributed by users rather than by the platforms themselves. This shows that contemporary design is not only about producing objects or visuals, but also about creating systems where interaction and participation can happen continuously.

The text also explains how participatory design connects to earlier theoretical and artistic ideas. Armstrong refers to Umberto Eco's concept of the "open work," which suggests that meaning is not completely fixed by the creator but is shaped through audience interpretation (Eco, 1989). This idea strongly connects to participatory design because users actively influence the final meaning of the experience through interaction and contribution. Armstrong also mentions movements such as Fluxus, which encouraged audience participation and blurred the boundary between creator and viewer long before digital technology made participation more widespread.

Another important point raised in the text is how participatory systems challenge traditional ideas of authorship and ownership. When many people contribute to the same system, it becomes more difficult to define who owns the final outcome. Although users provide content, they often lose control over how their contributions are used or distributed. This creates ethical questions surrounding creative labour, fairness, and intellectual property within participatory platforms.

Personally, this reading changed the way I think about design. Before reading the text, I mainly viewed design as the creation of finished outcomes controlled by the designer. Armstrong's argument instead presents design as something much more dynamic and collaborative, where meaning develops through interaction between systems and users. It also made me reflect on how many digital experiences today depend on participation in order to function. Overall, the text highlights how contemporary design is increasingly shaped by interaction, contribution, and shared participation rather than fixed control.

CTS Essay n.2

Case study: Social media

Social media reflect participatory design because users are not passive audiences, but active contributors who create , share , remix and shape content online.
socials are a very strong example of participatory design because as i said, users do not only consume content, but instead they actively create, share remix and interact with other media online everyday, and this connectss to helen arstrong's idea that participatory design changes the relationship between designer and users by allowing people to contribute directly to the final experience

Helen explains that traditional design is often based on control, where designers decide exactly how information is presented and how audiences should experience it, however social media works differently because these platforms depends heavly on user participation and interaction, without user creating content, posting videos commenting etc..., most social emdia platforms would't work properly.

Platforms such as TikTok, Instagram, YouTube, and Reddit are all built around participation. Users are constantly contributing content and shaping trends online. For example, on TikTok people often reuse the same sounds or video formats, but reinterpret them in their own way. This creates a collaborative system where content evolves through participation instead of being created by one single author.
This links strongly to Armstrong's idea that participatory design creates "openended systems." Social media platforms are never completely finished because users constantly reshape the experience through interaction. Meaning is created collectively by millions of people participating online rather than by one creator controlling everything.

Another important point is that social media has blurred the line between creators and audiences. In traditional media such as television, people mostly consumed content passively. On social media, however, users can become creators themselves very easily. Someone can watch content one moment and upload their own content the next. This reflects Armstrong's argument that participation allows audiences to become active contributors instead of passive viewers.

Social media also shows some of the problems connected to participatory design. Armstrong mentions issues surrounding ownership and control, and these problems are very visible online. Even though users create most of the content, platforms still control algorithms and visibility. This means users participate heavily, but they do not fully control how their content is distributed or used.

Overall, social media is a very good example of participatory design because it transforms users from passive audiences into active participants. People are constantly creating, interacting, and shaping online experiences together. Armstrong's text helps explain how participation has become central to digital culture today, where design is no longer only about delivering finished products, but about creating systems where interaction can continuously happen.

CTS Essay n.3

Participatory design and embodied interaction both explore how people actively engage with systems instead of passively receiving information. Helen Armstrong's What is Participatory Design? focuses on how users contribute to content and meaning, while Paul Dourish's Where the Action Is expands this idea by explaining how interaction is shaped through physical activity, movement, and experience within the real world. looking at these tw otexts together, i noticed how they create a broader undersatnding of interaction design as something based on participation, actiomn and human experience rather than control and passive observation

Armstrong explains that traditional design is often centred around control, where designers decide exactly how people should experience something. Participatory design challenges this idea by allowing users to contribute directly to the final outcome. Instead of users simply consuming finished content, they become active participants within open-ended systems (Armstrong, 2009). This changes the relationship between designers and audiences, as meaning is now shaped collectively through interaction and contribution.

Dourish develops a similar argument through the idea of embodied interaction. He argues that interaction does not only happen mentally through screens or interfaces, but physically through actions performed in the world around us. According to Dourish, meaning emerges through activity, movement, and social interaction rather than existing separately from experience (Dourish, 2001, p. 126). This supports Armstrong's idea because both authors suggest that engagement becomes stronger when people actively participate rather than observe passively.

One important connection between the two texts is the rejection of passive users. Armstrong describes how participatory systems encourage people to create, contribute, and interact directly with content. Similarly, Dourish explains that interaction should be understood through human practice and behaviour rather than only through technology itself (Dourish, 2001, p. 143). In both texts, users are seen as active participants who shape experiences through their actions.
Another important similarity is the idea that interaction is social and collaborative. Armstrong discusses how digital platforms and participatory systems allow large groups of people to collectively create content and meaning. Dourish also highlights how interaction often happens within social environments, where actions gain meaning through shared understanding and communication with others (Dourish, 2001). This suggests that interaction design is not only about interfaces or visual communication, but also about how people behave, move, and collaborate within systems.

These ideas are especially relevant today because many digital experiences are built around participation and embodied interaction. Social media, gaming, virtual reality, and interactive installations all rely on users actively engaging with systems rather than simply consuming information. Modern design increasingly focuses on creating experiences where people can explore, contribute, and interact naturally.
Overall, Armstrong and Dourish both argue that interaction should be understood as active participation rather than passive consumption. Armstrong focuses more on contribution and collaborative creation, while Dourish focuses on the physical and embodied nature of interaction. Together, the two texts create a strong contextual framing for understanding modern interaction design as something shaped through participation, action, and lived experience rather than fixed control from designers alone.

CTS Essay n.4

Helen Armstrong's What is Participatory Design? connects a lot to with many of the behaviours i observed inside the exhibition. Helen explains how participatory design changes the relationship between designers and audiences by moving people away from passive observation towards interaction. This connected well with my own observations surrounding tactile interaction and visitor engagement especially through button based artefacts at the welcome collection.

At the beginning of the project, my research focused more broadly on trust and identity within exhibitions. However, through repeated observations, I started noticing that the strongest behavioural patterns were connected more to participation and physical interaction. Interactive artefacts such as the Transparent Woman and the Jukebox encouraged visitors to actively engage through pressing buttons, replaying interactions, experimenting with controls, and interacting socially with others around them. Unlike nearby passive artefacts, these installations transformed visitors from observers into active participants within the exhibition environment.

Armstrong explains that participatory design creates open ended systems where users contribute to the final experience rather than simply receiving a finished outcome. This became visible during my observations. Visitors interacting with the Transparent Woman often pressed multiple buttons repeatedly to compare different illuminated organs and explore different outcomes. Similarly, the Jukebox encouraged visitors to select songs together, observe each other, dance, and repeatedly return to the interaction. In several situations, visitors appeared more curious and comfortable experimenting with tactile artefacts compared to nearby static displays.

These behaviours reflected Armstrong's argument that participation changes how people experience design. The installations were experienced through action, experimentation, and physical engagement rather than passive viewing alone. In many cases, visitors remained around the interactive artefacts for significantly longer periods compared to nearby non interactive displays. This suggested that tactile participation was influencing curiosity, replay behaviour, experimentation, and engagement within the exhibition space.

At the same time, my project also revealed some limitations surrounding participation. Although interactive artefacts generated stronger behavioural engagement, participation did not always guarantee deeper reflection or understanding. Some visitors appeared more focused on replaying interactions or experimenting with buttons than engaging critically with the exhibition content itself. This became an important reflection within my investigation because it suggested that participation can sometimes prioritise stimulation and entertainment over deeper interpretation.

Overall, Armstrong's ideas helped contextualise many of the behavioural patterns observed throughout my project. Through observation, synthesis, and reflection, my investigation showed that button based artefacts can encourage active participation, curiosity, experimentation, and social interaction within exhibition environments rather than passive observation alone.

Bibliography

O'Leary, Z. (2021) The Essential Guide to Doing Your Research Project.
Booth, W. et al. (2016) The Craft of Research.

Anon, (2019). The Importance of An Interactive Museum Exhibit | PLB Projects Ltd. [online] Available at: https://plbltd.com/the-importance-of-an-interactive-museum-exhibit/.

Arts Professional. (2025a). Museums' ad-hoc funding for learning activities jeopardising youth engagement - Arts Professional. [online] Available at: https://www.artsprofessional.co.uk/news/museums-ad-hoc-funding-for-learning-activities-jeopardising-youth-engagement.

Arts Professional. (2025b). Museums' ad-hoc funding for learning activities jeopardising youth engagement - Arts Professional. [online] Available at: https://www.artsprofessional.co.uk/news/museums-ad-hoc-funding-for-learning-activities-jeopardising-youth-engagement.

Berta, M., Melese, K.B. and Dessie, A. (2025). The effect of interactive technology on enriching visitor experiences: Evidence from the science museum, Addis Ababa, Ethiopia. Computers in Human Behavior Reports, [online] 20, p.100869. doi:https://doi.org/10.1016/j.chbr.2025.100869.

Booth, W.C., Colomb, G.G., Williams, J.M. and Al, E. (2016). The craft of research. Chicago, Ill.: University Of Chicago Press.

FIWebteam (2017). Radical Change: Participatory Exhibition Design. [online] Museum-iD. Available at: https://museum-id.com/radical-change-participatory-exhibition-design-by-silvia-filippini-fantoni/.

IDEO (2015). The Field Guide to human-centered design. 1st ed. San Francisco, Calif: Design Kit.

eanhochu.com. (2026). Jean Ho Chu || Tangible Interaction Design. [online] Available at: https://www.jeanhochu.com/mapping-place.html [Accessed 2 Jun. 2026].

McEwan, T., Gulliksen, J. and Benyon, D. eds., (2006). People and Computers XIX — The Bigger Picture. [online] London: Springer London. doi:https://doi.org/10.1007/1-84628-249-7.

Museumsassociation.org. (2009). Home | Museums Association. [online] Available at: https://www.museumsassociation.org/.

O'Leary, Z. (2021). The Essential Guide to Doing Your Research Project. 4th ed. S.L.: Sage Publications.

Paiva, A.C.R., Rui Prada and Picard, R.W. (2007). Affective Computing and Intelligent Interaction : Second International Conference, ACII 2007 Lisbon, Portugal, September 12-14, 2007 Proceedings. Berlin, Heidelberg: Springer Berlin Heidelberg.

Public perceptions of -and attitudes to - the purposes of museums in society A report prepared by BritainThinks for Museums Association.

(2013). Available at: https://www.museumsassociation.org/app/uploads/2020/06/03042013-britain-thinks.pdf.

Richardson, J. (2022). 6 museums successfully cementing relationships with their audiences. [online] MuseumNext. Available at: https://www.museumnext.com/article/6-museums-successfully-cementing-relationships-with-their-audiences/.

Stickdorn, M., Hormess, M., Lawrence, A. and Schneider, J. (2018a). This is service design doing applying service design thinking in the real world : a practioner's handbook. Sebastopol, CA: O'Reilly.

Stickdorn, M., Markus Edgar Hormess, Lawrence, A. and Schneider, J. (2018b). This Is Service Design Doing. 'O'Reilly Media, Inc.'